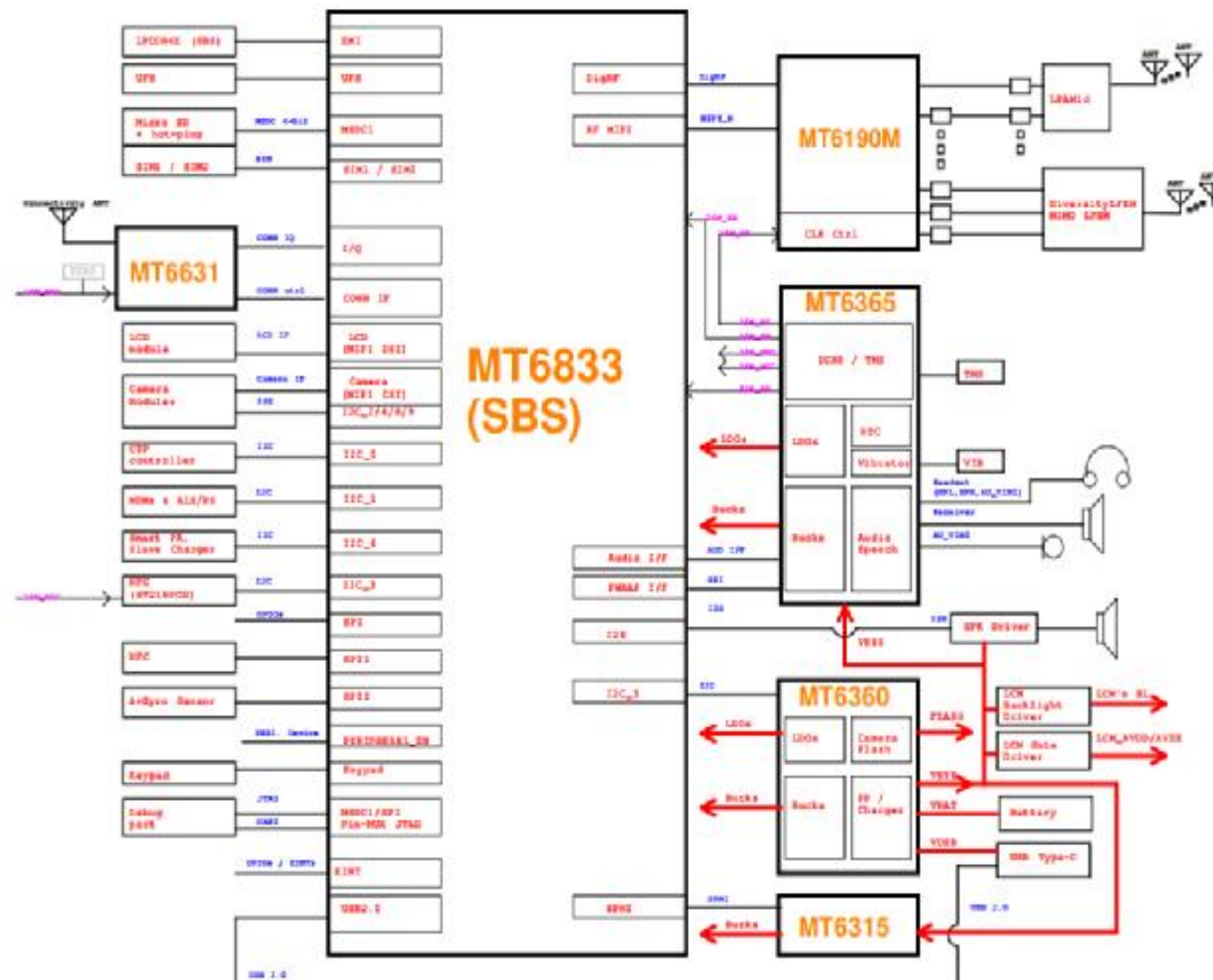


BB		RF	
01_Index			
02_Block_Diagram	32_CAMERA IF	39.Customer_RFI	65.GPS
03_I2C_ID_Overview	33_Flash LED	40.RF_Block	66.NFC_PN553
04_Change_Notice	34_FingerPrint	41.RF_Transceiver_MT6190_Power	
05_BB_POWER_PDN	35_SENSORS	42.RF_Transceiver_MT6190	
06_BB_POWER_PDN	36_SIM/SD IF	43.RF interface	
07_BB_POWER_IO	37_Testpoint/Shielding/GND	44.APT/ET_Module_1	
08_BB_1	38_Sub PCB IF / USB IF	45.APT/ET_Module_2	
09_BB_2		46.FBRX_SWITCH	
10_BB_3		47.LTE_LB_TRX	
11_BB_4		48.LTE_MHB_TRX	
12_BB_AUXADC_Thermal		49.MMBPA_ENDC	
13_POWER_MT6359-Buck		50.N41_TRX	
14_POWER_MT6359-LDO		51.N77/78/79_TRX	
15_POWER_MT6359-General		52.RF_LMH_PRX	
16_POWER_MT6359_Clock		53.RF_LMH_DRX	
17_POWER_MT6315		54.N78/79_RX and MIMO	
18_POWER_MT6315_VMD		55.Antenna_0_PRX_LB+MIMO_MHB	
19_POWER_MT6360_BUCK_LDO		56.Antenna_1_PRX_UHBCB+N41	
20_POWER_MT6360_General		57.Antenna_2_DRX_UHBCB+N41	
21_POWER_MT6360_Charger		58.Antenna_6_MIMO_UHBCB+N41	
22_POWER_EXT		59.Antenna_7_MIMO_UHBCB+N41	
23_Memory_UFS_LPDDR4X		60.Antenna_5_DRX_LMHB	
24_POWER_MT6359_AUDIO		61.Sar_sensor_SX9338IULTRT	
25_SPK		62.WCN_IC_Power_GND	
26_Receiver		63.WCN_RF0_ANT3	
27_MIC		64.WCN_RF1_ANT4	
28_Earphone			
29_Battery/Sidekey			
30_Charger_V600			
31_LCD/CTP IF			

02_Block_Diagram

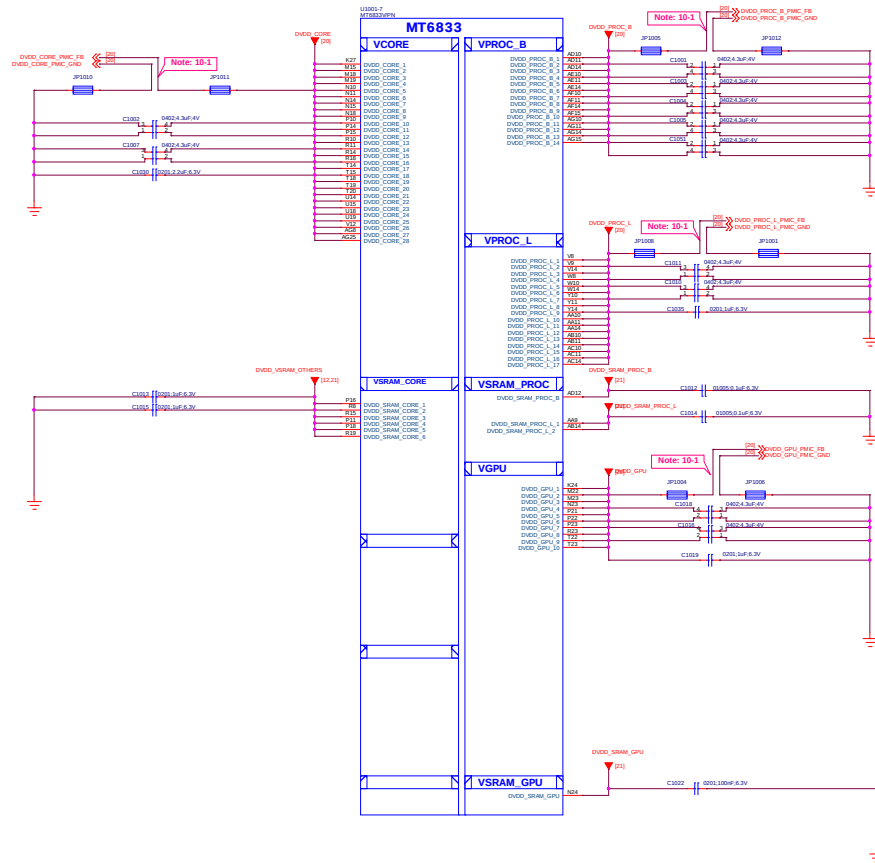


04_Change_Notice

Date	Version	Page	Change notice
20120.10.15			Changed from 5G-C HW
20120.10.16			1st release

Title		003_Change_Notice		REV: 003	
DOCUMENT NO:		98016_1_16_20230808_1440		Size: D	
DEPARTMENT:		BB		DESIGNER: huanghanyu	
					
Date:		Tuesday, August 15, 2023		Sheet 3 of 323	

10_BB_POWER_PDN

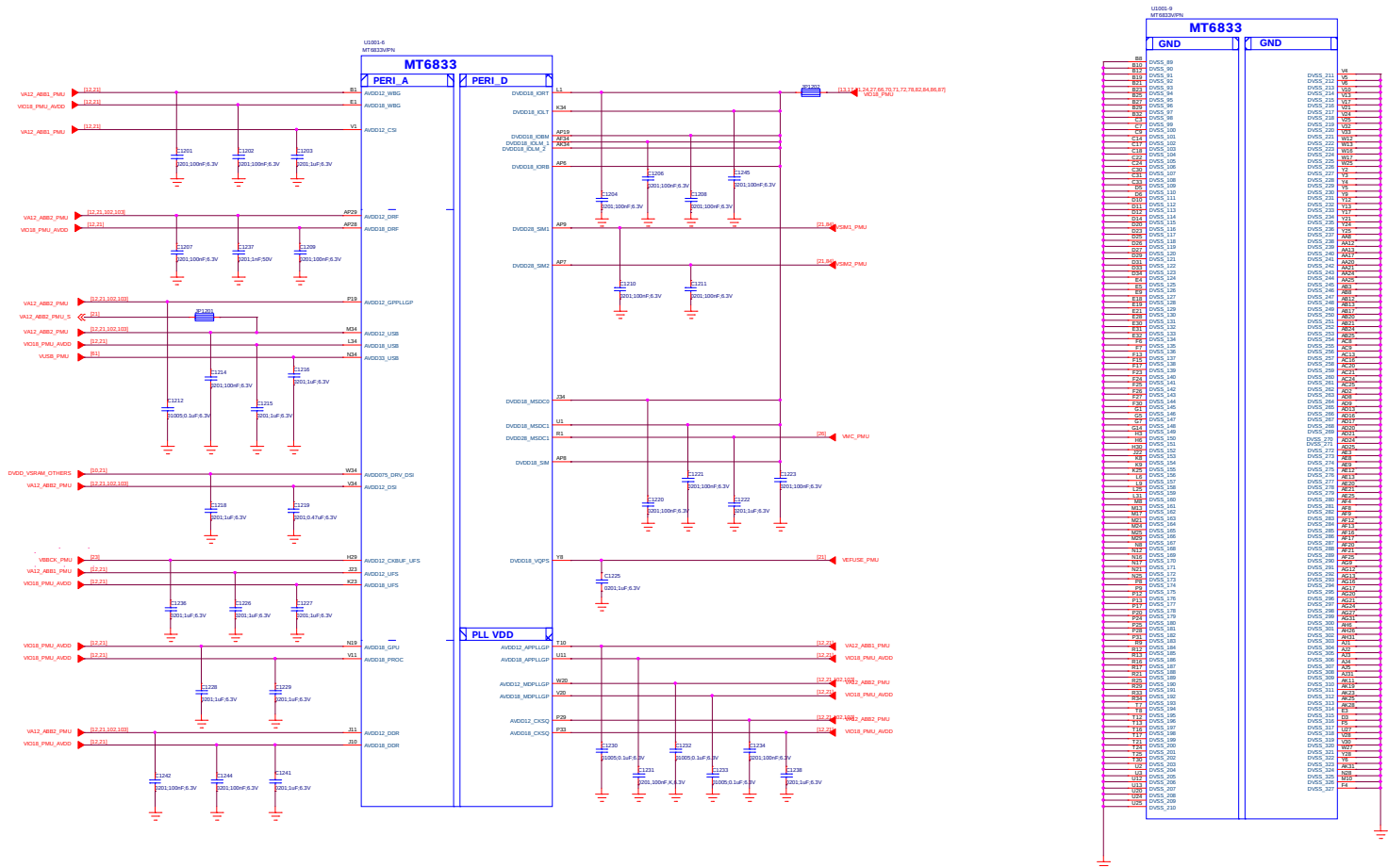


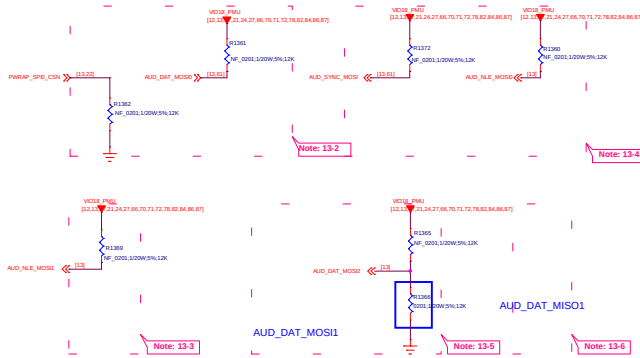
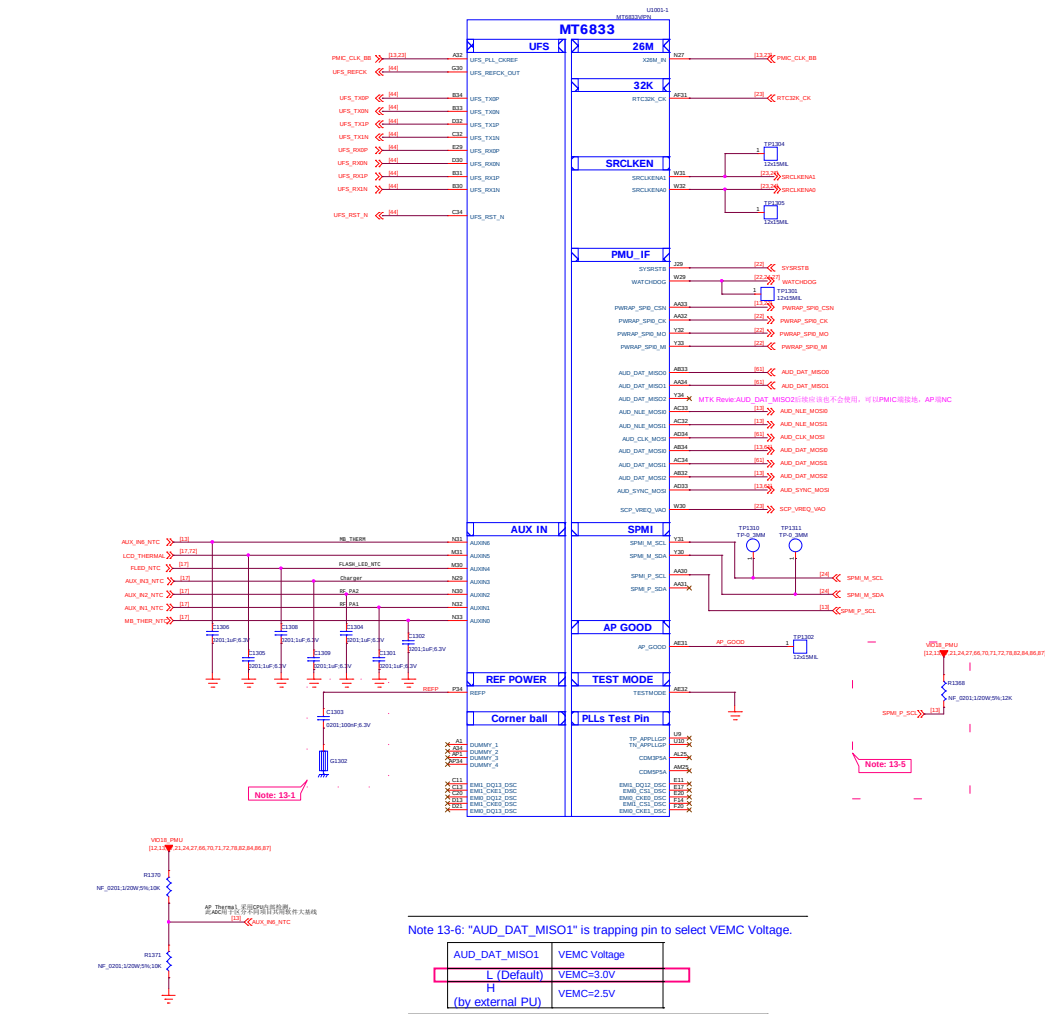
Schematic design notice of "10_BB_POWER_PDN" page.

Note 10-1: Differential pair of PMIC remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

Title		010_BB_POWER_PDN	REV: V10
DOCUMENT NO.:		90016_1_58_20210809_1440	Size D
DEPARTMENT:		BB	DESIGNER: huangshuang
			
Date:		Tuesday, August 10, 2021	Sheet 10 of 152

12_BB_POWER_IO





Schematic design notice:

Note 13-1: The load cap. have to be placed as close to REFP ball as possible.

Note 13-2: "PWRAP_SPI0_CSN" and "AUD_DAT_MOS0" pin features in trapping pin to enable JTAG.

PWRAP_SPI0_CSN	AUD_DAT_MOS0	AP JTAG	IO JTAG
H (Default)	L (Default)	N/A	N/A
H	H	SPI0+ENT8	N/A
L	L	SPI0+ENT8	ENT16/ENT17/ENT18/ENT19/ENT20
L (by external PD)	L	SPI0+ENT8	ENT16/ENT17/ENT18/ENT19/ENT20
L	H	MSDC1_CLK, MSDC1_CMD, MSDC1_DAT0, MSDC1_DAT1, MSDC1_DAT2	N/A

Note 13-3: "AUD_NLE_MOS1" features in trapping pin to enable JTAG over USB 2.0 port.

AUD_NLE_MOS1	USB JTAG
L (Default)	USB 2.0
H	DPI/DM output JTAG

Note 13-4: "AUD_SYNC_MOS1" and "AUD_NLE_MOS10" pin features in trapping pin to booting (eMMC/UFS).

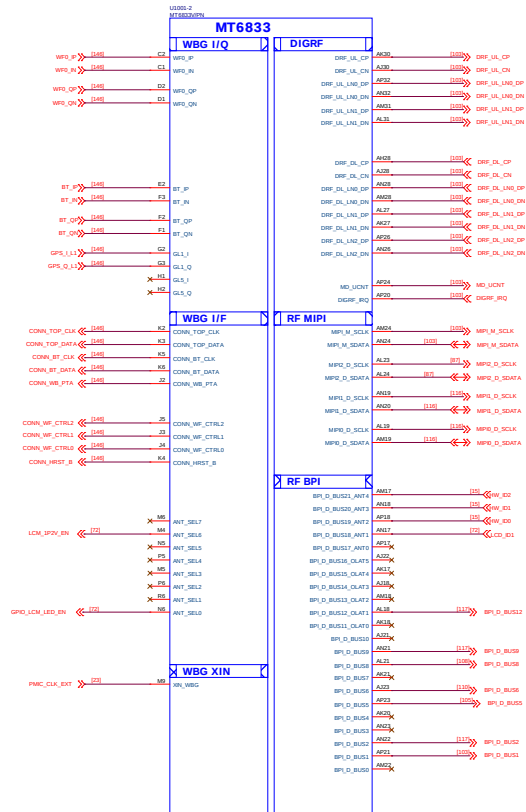
AUD_SYNC_MOS1	AUD_NLE_MOS10	Storage Booting
L (Default)	L (Default)	Only UFS boot
L	H	Only eMMC boot
H (by external PU)	L	Reserved
H (by external PU)	H (by external PU)	Reserved

Note 13-5: "AUD_DAT_MOS11" and "AUD_DAT_MOS12" pin features in trapping pin to enable DDR.

AUD_DAT_MOS11	AUD_DAT_MOS12	SPMI_P_SCL	DDR
L (Default)	L (Default)	L (Default)	LP4x-MCP-4266Mbps
L	H	L (Default)	LP4x-Discrete-4266Mbps
H (by external PU)	L	L (Default)	Reserved
H (by external PU)	H (by external PU)	L (Default)	Reserved

Note : SPMI_P_SCL for DDR 4266Mbps/3733Mbps Co-load, 'L' = 4266Mbps, 'H (by external PU)' = 3733Mbps

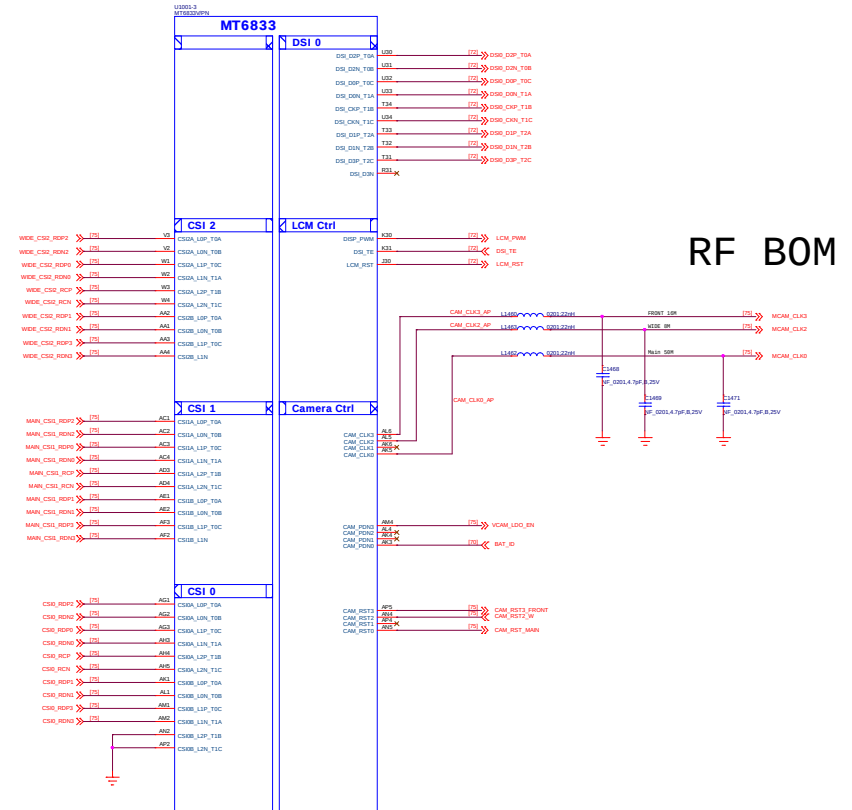
14_BB_2



WIDE 8M

Main 50M

Front 16M

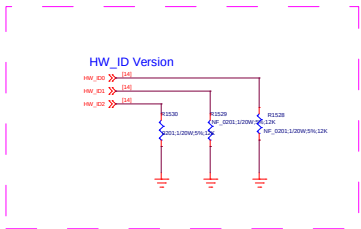
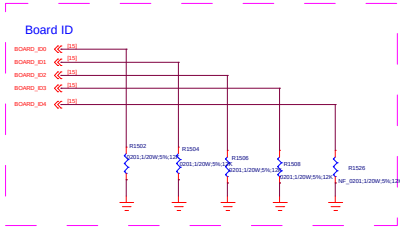
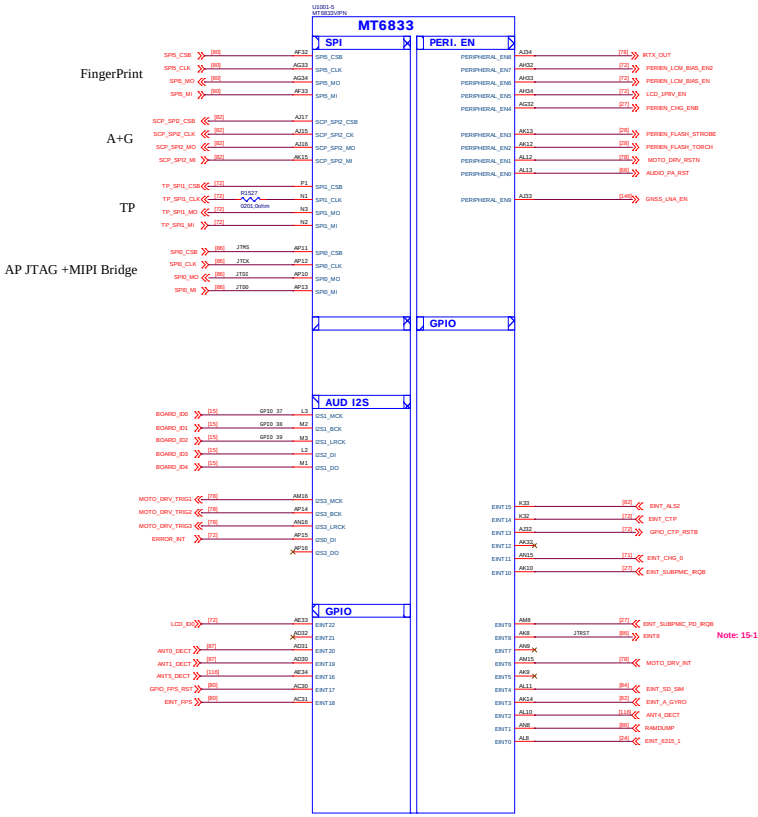
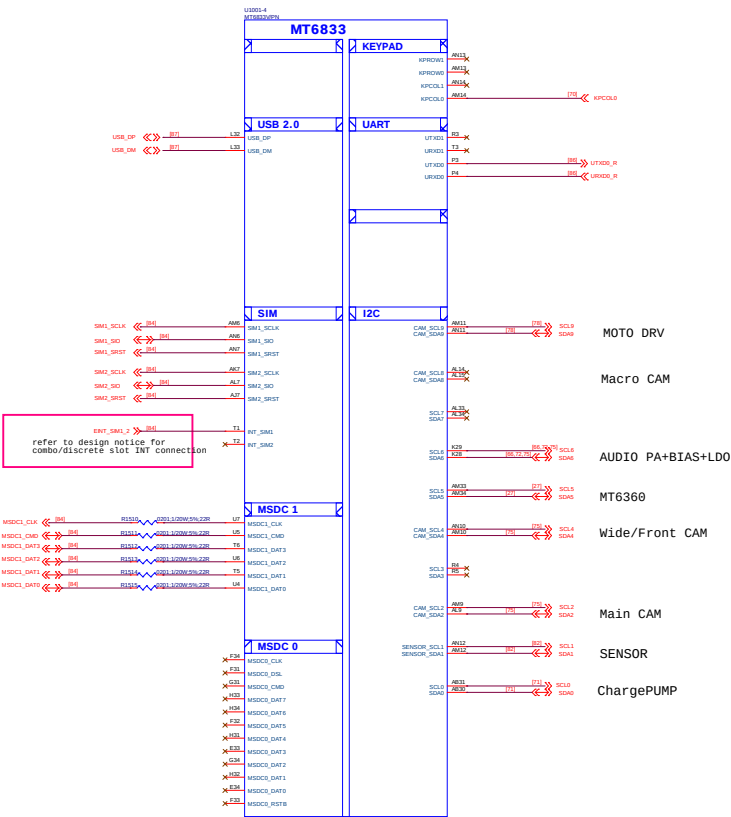


RF BOM

Title		014_BB_2_MIPI		REV: V1.0	
DOCUMENT NO.: 90018_1_16_2010000_1440				Size D	
DEPARTMENT: BB			DESIGNER: huanghongli		
					
Date: Tuesday, August 10, 2021			Sheet 14 of 152		

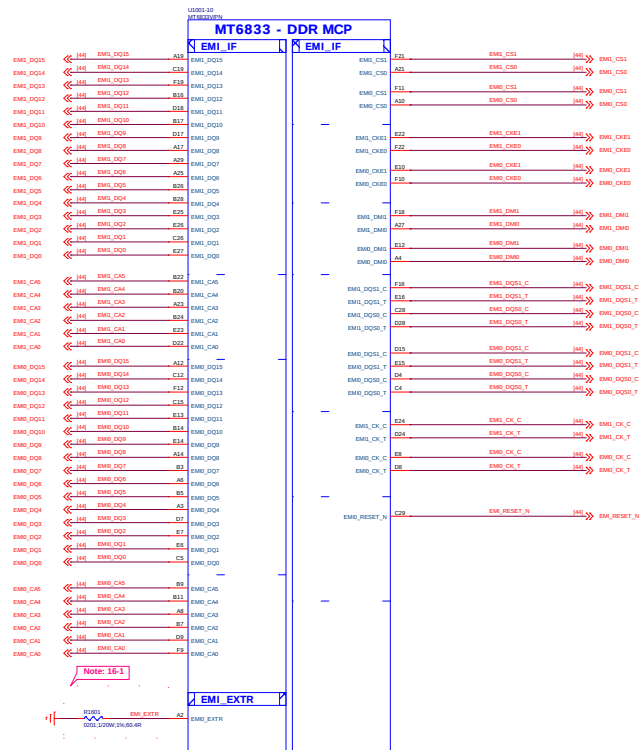
close the SoC

15_BB_3



HW Version	010	011	100	R1038	R1039	R1038
00017_1_00	0	0	0	SMT	SMT	SMT
00017_1_10	0	0	1	SMT	SMT	MT
00017_1_11	0	1	0	SMT	MT	SMT
00017_1_12	0	1	1	SMT	MT	MT
00017_1_13	1	0	0	MT	SMT	SMT
00017_1_14	1	0	1	MT	SMT	MT
00017_1_15	1	1	0	MT	MT	SMT
00017_1_16	1	1	1	MT	MT	MT

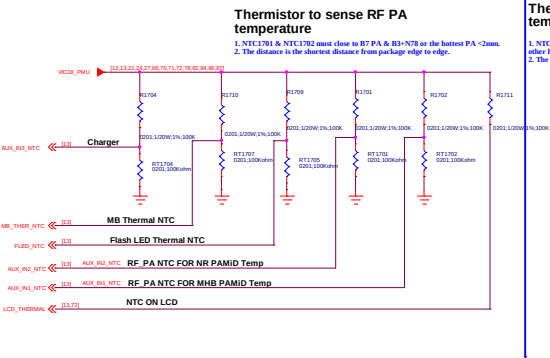
File	015_BB_3_MSC	REV: V00
DOCUMENT NO:	00014_1_16_0030000_1440	Size: D
DEPARTMENT:	BB	DESIGNER: huanghongen
DATE:	Tuesday, August 10, 2021	Sheet: 15 of 153



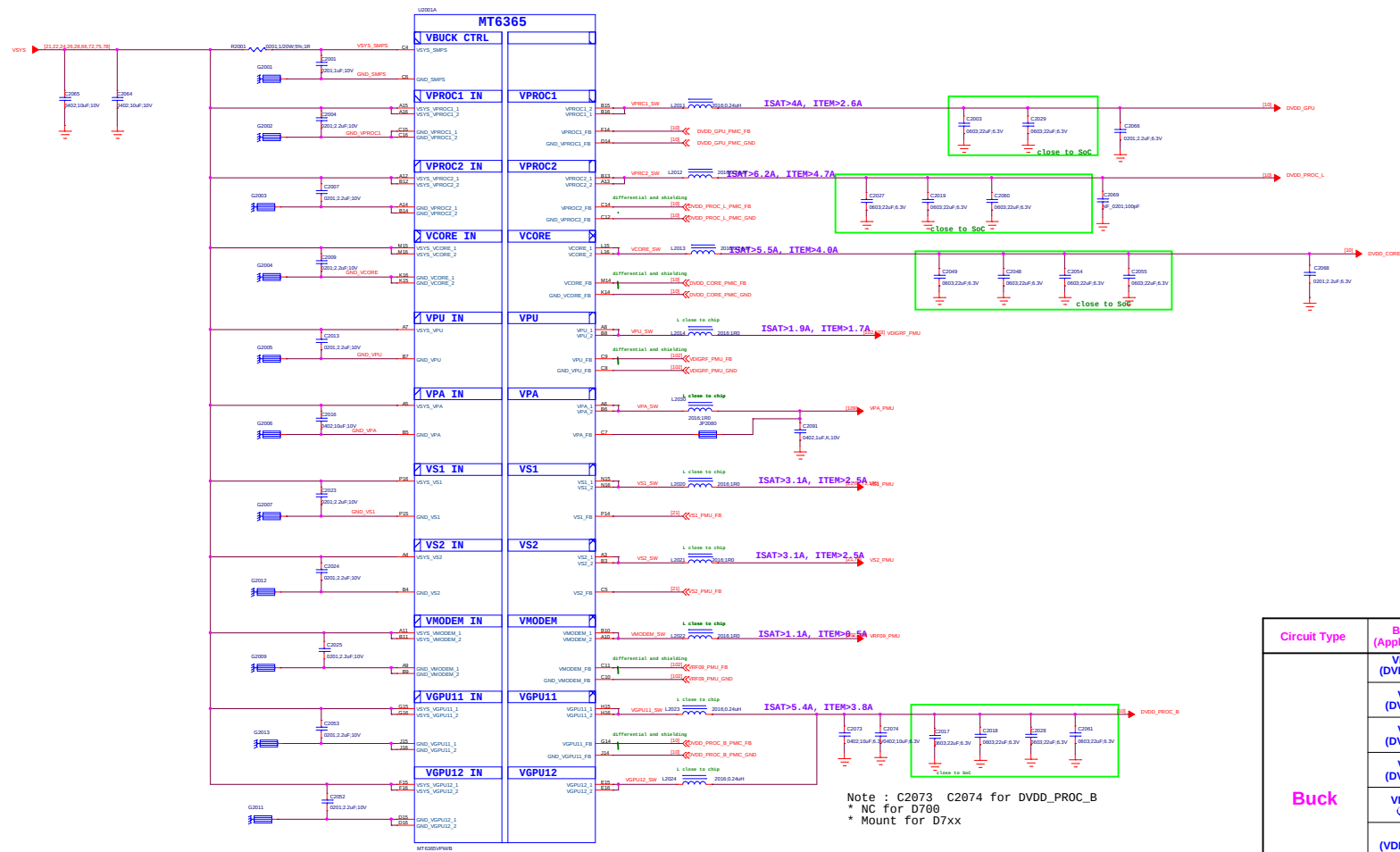
Schematic design notice of "16_BB_4 Interface" page:

Note 16-1: R4001 please select 60.4 ohm (1%) resistor

17_BB_AUXADC_Thermal

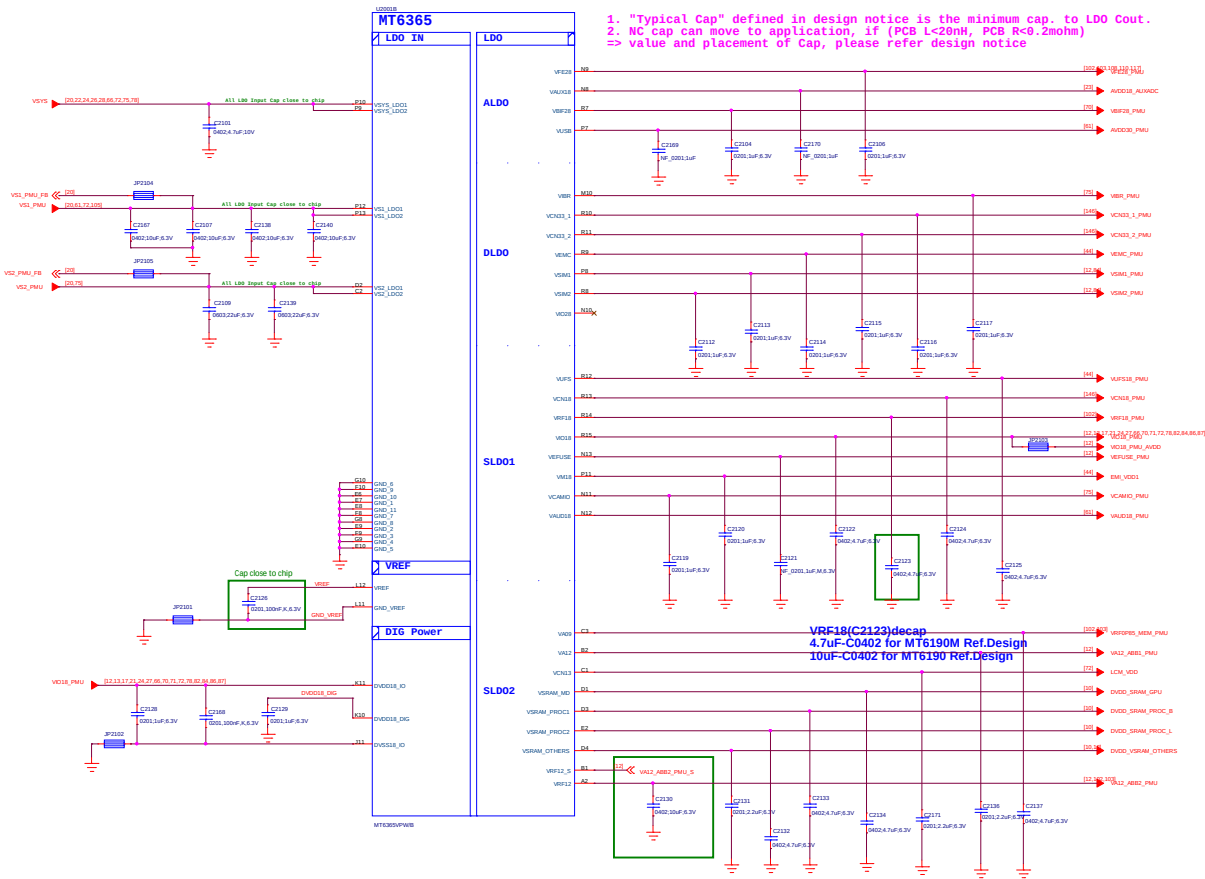


20_POWER_MT6365-Buck



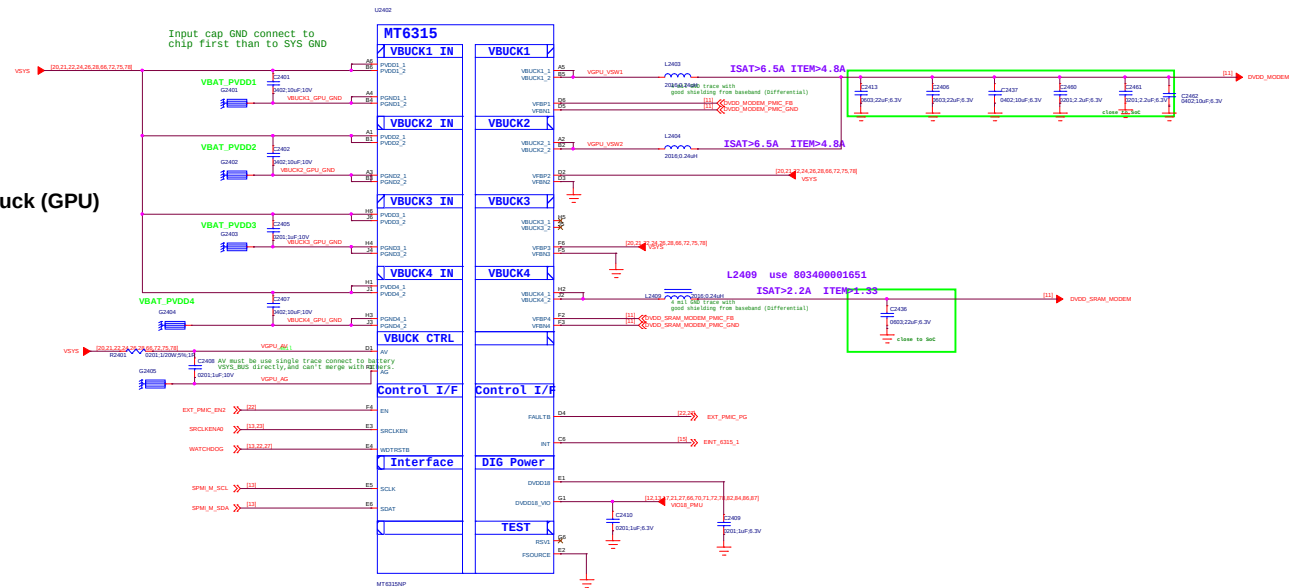
Circuit Type	BUCK Name (Application name)	Output Voltage Range(V)	BOOT Defalut	lout max(mA)
Buck	VPROC1 (DVDD_GPU)	0.40~1.19 (6.25mV/step)	ON(0.75)	4800
	VPROC2 (DVDD_PROC_L)	0.40~1.19 (6.25mV/step)	ON(0.75)	4800
	VGP11+12 (DVDD_PROC_B)	0.40~1.19 (6.25mV/step)	ON(0.75)	4800*2
	VCORE (DVDD_CORE)	0.40~1.3 (6.25mV/step)	ON(0.75)	4800
	VMODEM (VRF09_PMU)	0.50~1.10 (6.25mV/step)	OFF	4800
	VPU (VDIGRF_PMU)	0.40~1.19 (12.5.mV/step)	ON(0.7)	2400
	VS1 (VS1_PMU)	1.86~2.20 (12.5.mV/step)	ON(2.0)	2200
	VS2 (VS2_PMU)	1.20~1.50 (12.5.mV/step)	ON(1.35)	2500
	VPA (VPA_PMU)	0.50~3.40 (50mV/step)	OFF	1000

21_POWER_MT6365-LDO

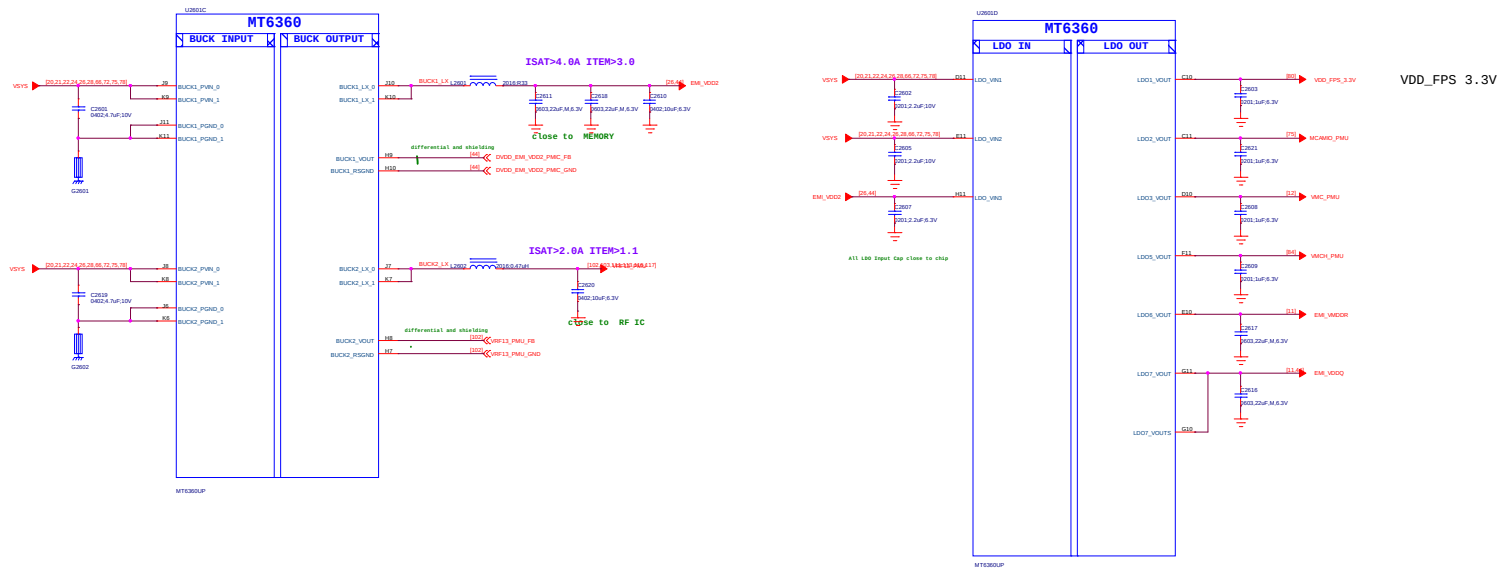


LDO Name	Output Voltage(V)	BOOT Defalut	out max(mA)	Expected use
VEF28	2.80	OFF(2.8)	200	RFFE
VUSB	3.07	ON(3.07)	200	USB
VAUX18	1.84	ON(1.84)	50	AUXADC
VXO22	2.24	ON(2.24)	25	DCXO
VRFCK	1.4@boot 1.6@work	ON(1.4)	10	DCXO
VBIF28	2.80	OFF(2.8)	50	Battery Interface
VRTC	2.80	ON(2.8)	2	RTC
DVDD18_DIG	1.80	ON(1.8)	10	PMIC Digital
VSIM1	1.7/1.8/1.86/2.76/3.0/3.1	OFF(1.86)	200	SIM
VSIM2	1.7/1.8/1.86/2.76/3.0/3.1	OFF(1.86)	200	SIM
VCN33_1	3.3/3.4/3.5/3.6	OFF(3.3)	800	Connectivity
VCN33_2	3.3/3.4/3.5/3.6	OFF(3.3)	800	Connectivity
VEMC	2.55/2.9/3.0/3.3	ON(3.0)	800	eMMC/UFS
VIO28	2.8/2.9/3.0/3.1/3.2/3.3	OFF(2.8)	200	IO&Sensor
VIBR	2.7/2.8/3/3.3	OFF(2.8)	200	Vibrator
VEFUSE	1.80	OFF(1.8)	300	EFUSE
VAUD18	1.80	ON(1.8)	300	Audio
VCAMIO	1.80	OFF(1.8)	300	Camera IO
VM18	1.80	ON(1.8)	300	DRAM
VRF18	1.80	OFF(1.8)	450	RF
VIO18	1.80	ON(1.8)	600	IO&Sensor
VCN18	1.80	OFF(1.8)	1200	Connectivity
VUFS	1.86	ON(1.86)	1200	eMMC/UFS
VBBCK	1.24	ON(1.24)	10	DCXO
VA09	0.85	ON(0.85)	300	DIGRF SRAM (VRF0P85 MEM_PMU)
VA12	1.2	ON(1.2)	300	AP Analog Module
VCN13	1.3	OFF(1.3)	350	Connectivity
VSRAM_PROC1	0.60-1.2 (6.25mV/step)	ON(0.85)	600	CPUB SRAM
VSRAM_PROC2	0.60-1.2 (6.25mV/step)	ON(0.85)	600	CPUL SRAM
VSRAM_OTHERS	0.60-1.2 (6.25mV/step)	ON(0.75)	600	OTHERS SRAM
VSRAM_MD	0.60-1.2 (6.25mV/step)	ON(0.85)	600	GPU SRAM
VRF12	1.2	ON(1.2)	800	AP Analog Module

MT6315 4-Phase Buck (GPU)

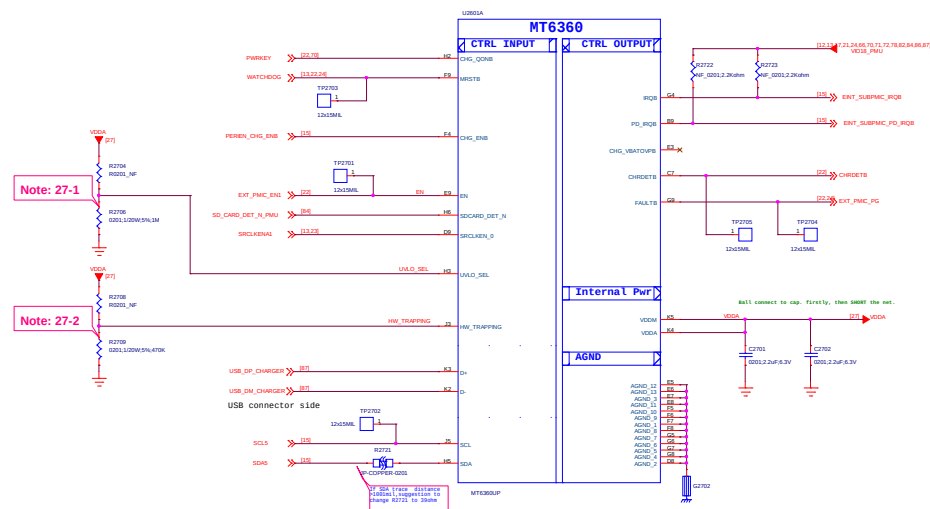


26_POWER_MT6360_BUCK_LDO



Circuit Type	Name	Output Voltage Range(V)	BOOT Defalut	Iout max(mA)	Expected use
Buck	BUCK1	0.3~1.3 (5mV/step)	ON(1.125)	3000	VDRAM1
	BUCK2	0.3~1.3 (5mV/step)	ON(0.75)	3000	VSRAM_CORE+ADSP
LDO	LDO1	1.8/2/2.1/2.5/2.7/ 2.8/2.9/3/3.1/3.3	OFF(1.8)	150	Fingerprint
	LDO2	1.8/2/2.1/2.5/2.7/ 2.8/2.9/3/3.1/3.3	OFF(1.8)	200	CAM AVDD
	LDO3	1.8/2.9/3/3.3	OFF(3.0)	200	SD Card
	LDO5	2.9/3/3.3	OFF(3.0)	800	SD Card
	LDO6	0.75	ON(0.75)	300	EMI_VMDDR
	LDO7	0.6	ON(0.6)	600	EMI_VDDQ

27_POWER_MT6360_General



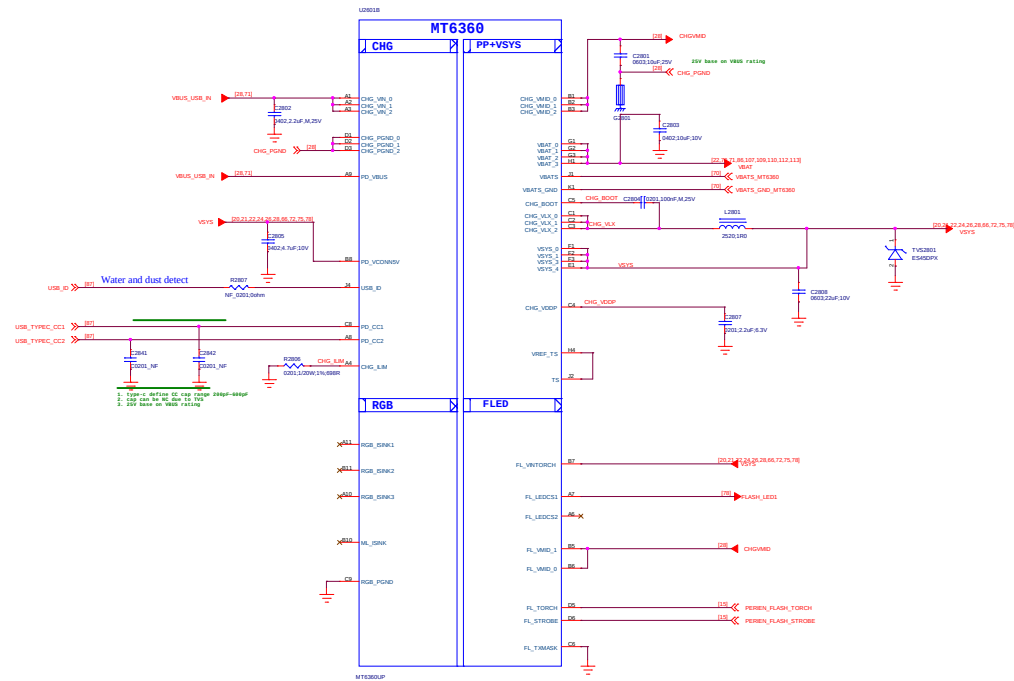
Note 27-1: UVLO_SEL pin is define difference UVLO power configuration.The power configuration is used to Ruvlosel (R2706) to select.

Note 27-2: HW_TRAPPING is define difference power configuration.The power configuration is used to Rhw_trapping (R2709&R2704) to select.

REG. PMU0x04[7:5]: HW_SYSVU_SEL[2:0]	RVU0_SEL_Range[D]		
	Min	Typ	Max
000 (2.8V)	Short to VDDA		
001 (2.9V)	936K	1.8M	---
010 (3.0V)	234K	430K	663K
011 (3.1V)	58K	100K	165K
100 (3.2V)	14K	28K	41K
101-111 (3.3V)	---	7.5K	10K

REG. PMU0x98[2:0] : HW_TRAPPING[2:0]	Rhw. trapping Range(C)			BUCK1	BUCK2	LDO6	LDO7
	Min	Typ	Max				
000	Short to VDDA			0.725V	1.125V	0.75V	0.6V
001	936K	1.8M	---	0.725V	1.125V	X	1.8V
010 (Default)	234K	430K	663K	1.125V	0.725V	0.75V	0.6V
011	58K	100K	165K	1.125V	0.725V	X	1.8V
100	14K	28K	41K	0.725V	1.225V	X	X
101	3.8K	7.5K	10K	1.225V	0.725V	X	X
110	0.96K	1.6K	2.4K	0.725V	1.125V	X	X
111	---	422	603	0.725V	0.725V	X	X

28_POWER_MT6360_Charger

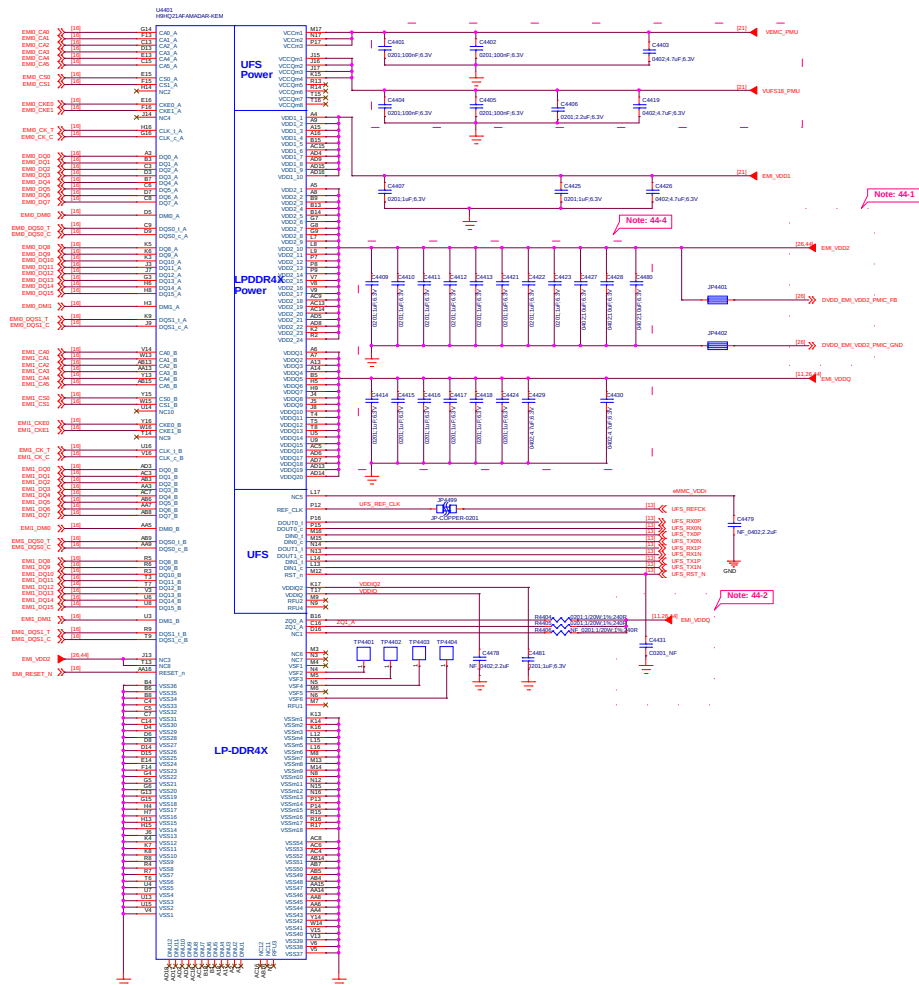


Title		028_POWER_MIT6360_Chapter		Ver: V3.0	
DOCUMENT NO.:				90016 1 16 20210809_1440	
DEPARTMENT:				DESIGNER: huanghonglai	
					
Date:		Tuesday, August 10, 2021		Sheet 28 of 152	

Title		029_Reserved	REV: 000
DOCUMENT NO:		98024_1_56_20230809_3440	Size D
DEPARTMENT:		BB	DESIGNER: huanghanyu
Date:		Tuesday, August 15, 2023	Sheet 29 of 323



44_Memory_UFS_LPDDR4X



Schematic design notice of "44_Memory_UFS_LPDDR4X" page.

Note 44-1: Please refer to power supply related page select LDO7_VOUT / BUCK1_LX output voltage properly for LPDDR4X

Note 44-2: DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ.

Note 44-3: Please refer to uMCP vendor's datasheet or MTK common design notice to get the recommendation bypass cap. value for VCC/VCCQ/VDDI power domains of UFS.

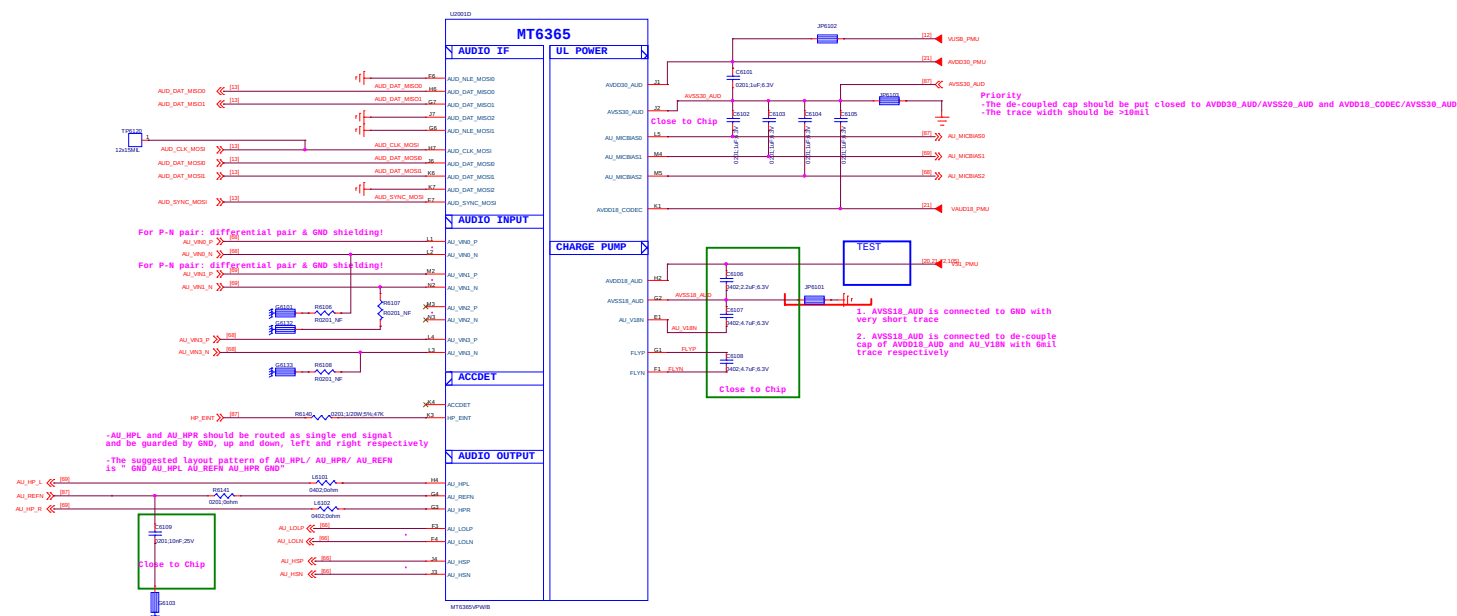
Note 44-4: VDD2 VDDQ decoupling cap: closed to DRAM ball.
For other cap for PMIC >10uF, at PMIC page;
please also refer to MMD and layout guide for placement.

02_MT6382_V0.2

[illegible]

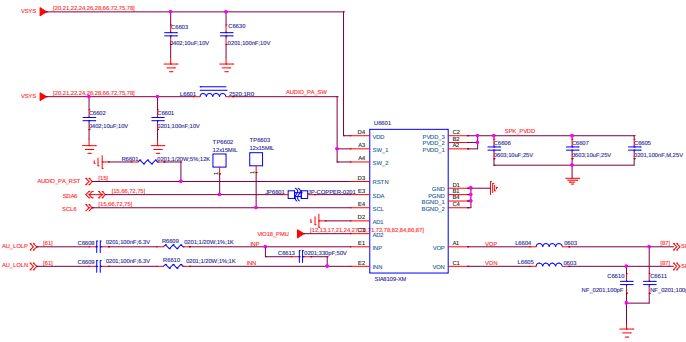
Title		060_MT6382_MPI Bridge	REV: V1.0
DOCUMENT NO.:		060216_1_36_20210809_1A40	Size: D10
DEPARTMENT:		EE	DESIGNER: huanghengfei
			
Date: Tuesday, August 10, 2021		Sheet 80 of 1	

61_POWER_MT6365_AUDIO

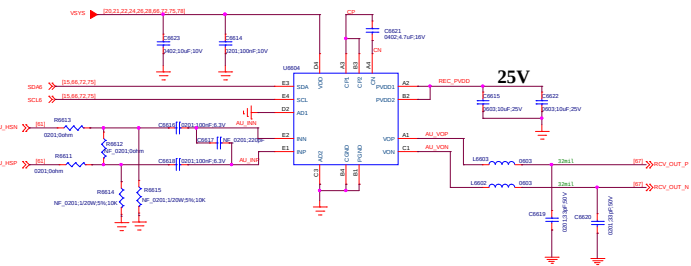


Title		061_PERI_6365_AUDIO_PM_C516	
DOCUMENT NO.:		00016_1_16_20210009_1440	Size D
DEPARTMENT:		BB	DESIGNER: huanghong
			
Date:		Tuesday, August 10, 2021	Sheet 61 of

FOR SPK



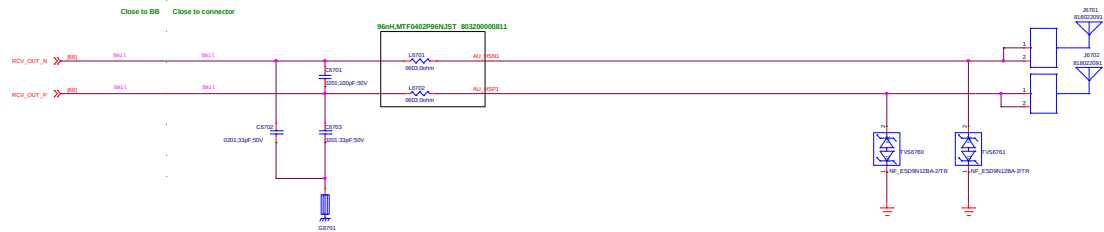
FOR REC



兼容艾为AW87389

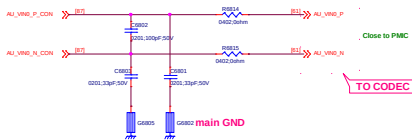
067_Receiver

Receiver

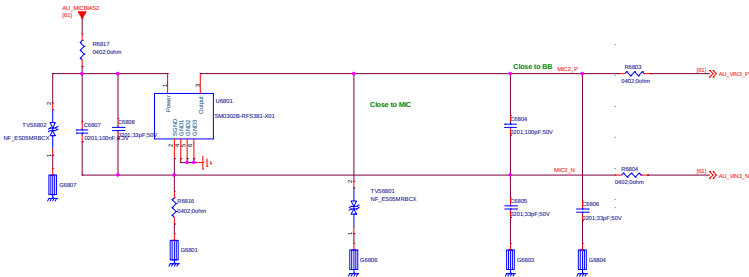


Title067_RECEIVER		REV: V02
DOCUMENT NO.: 06024_1_M_20210609_1408		SW: D
DEPARTMENT: BB	DESIGNER: huangshuang	
Date: Tuesday, August 10, 2021		Sheet 07 of 102

MAIN MIC



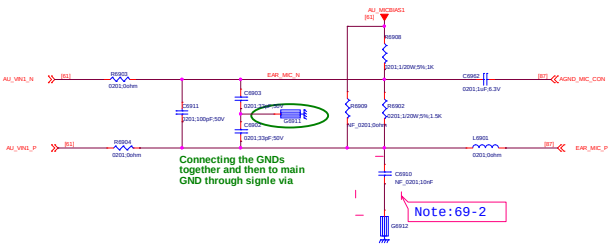
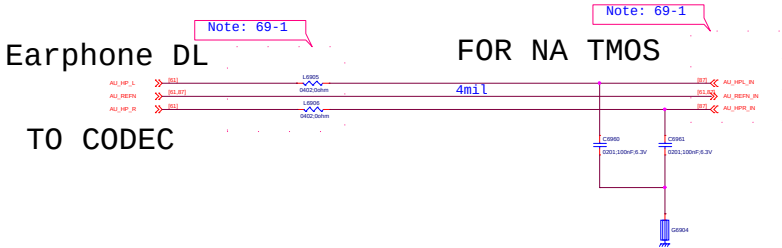
AUX MIC



Title068_MIC		REV: 000
DOCUMENT NO: 98024_1_56_20230809_1440		Size D
DEPARTMENT: BB	DESIGNER: huanghanyu	
Date: Tuesday, August 15, 2023		Sheet 00 of 00

069_Earphone

Earphone MIC



Schematic design notice of "62_PERI_AUDIO_IO" page.

Note 69-1: Part # of BEAD6202, BEAD6203, BEAD6204 and BEAD6205 needs changed to "BLM18BD102SN1" for high THD performance (-90dB) but this BOM change will results in FM RSSI 10dB degraded .

Note 69-2: Reserved Cap C6910 for CS/RS test, please double check multi-key function when used

Note 4-3:

Earphone Jack	Earphone Jack
8 Pin	8 Pin
1000P	1000P

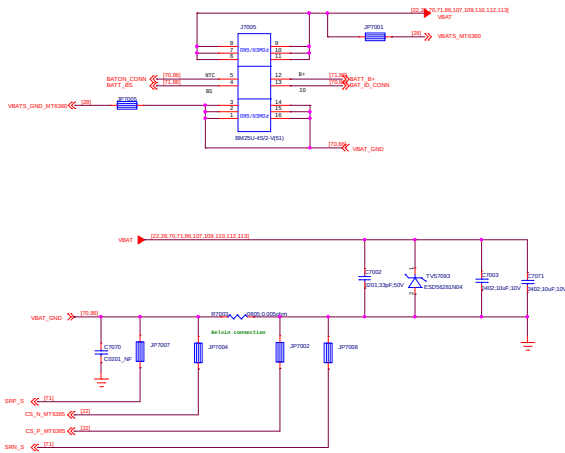
Note 4-4: 1. Need do SW configuration for DCC mode
2. Analog input impedance is as below, please customer check the detail requirement with mic vendor
if the mic used has special requirement on the input impedance of analog input

RT6357	UL Mode	REN	TYP	MAX	Unit
C	DCC	-	-	10p	F

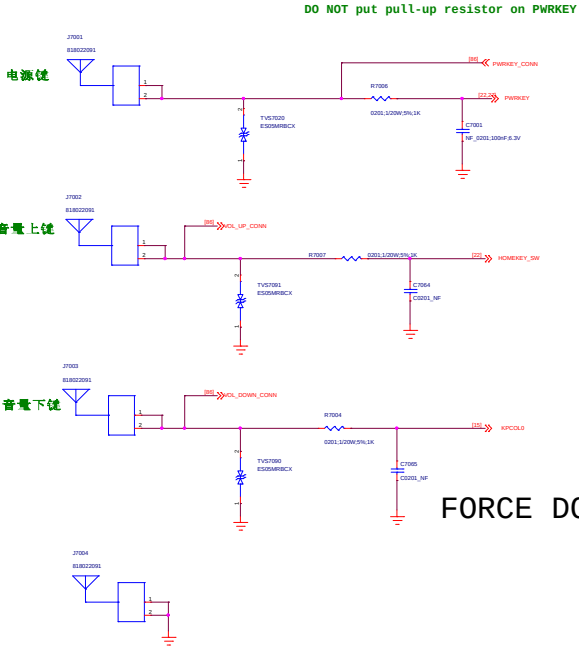
Note 4-5: Please select R6231 with 0402 size

070_Battery/Sidekey

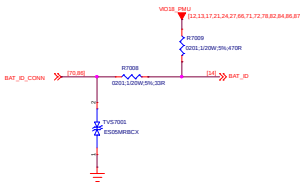
BATTERY CONNECTOR



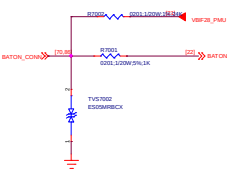
POWER KEY /VOL KEY



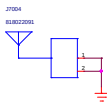
BAT ID



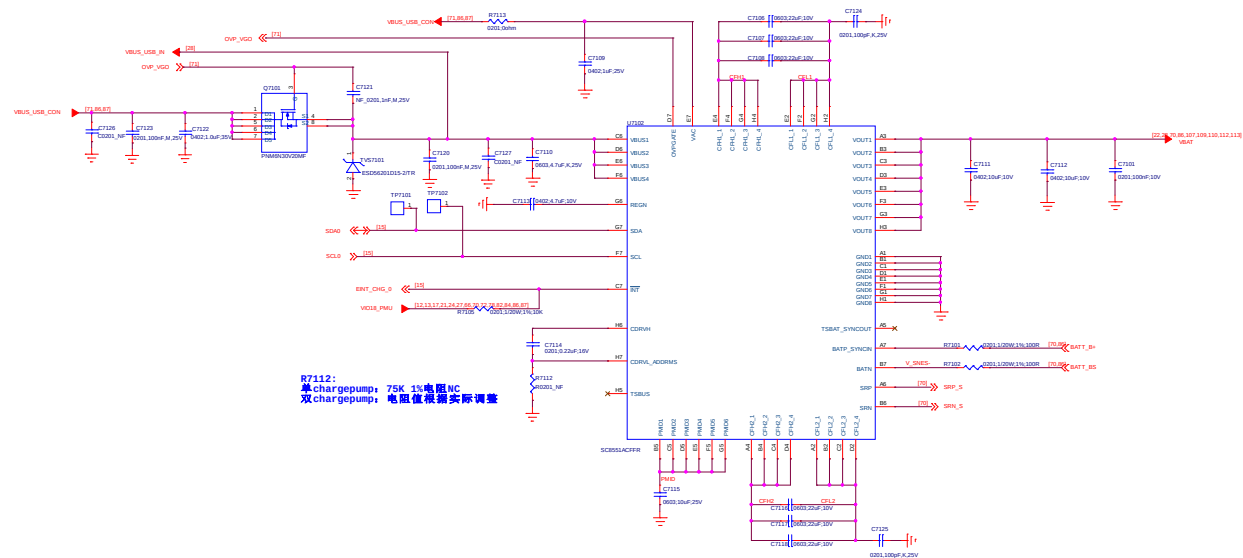
Thermal



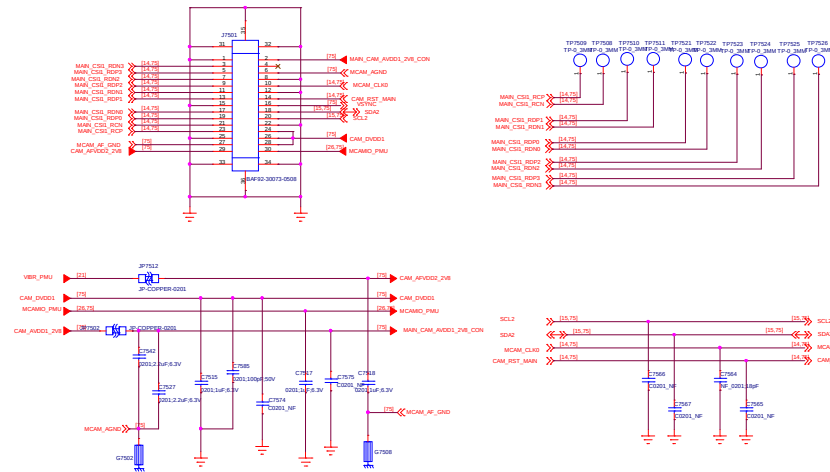
FORCE DOWNLOAD



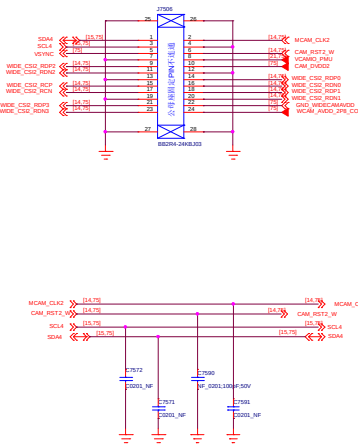
071_Smart Cap Divider Charger



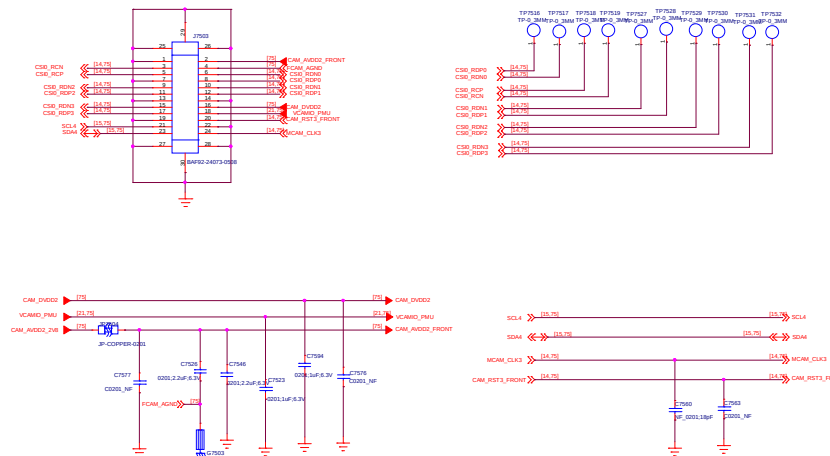
075_CAMERA IF Main Camera 50M



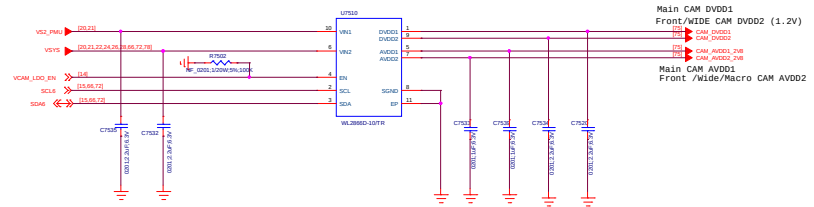
Wide Camera 8M



Front Camera 16M



LDO for Camera



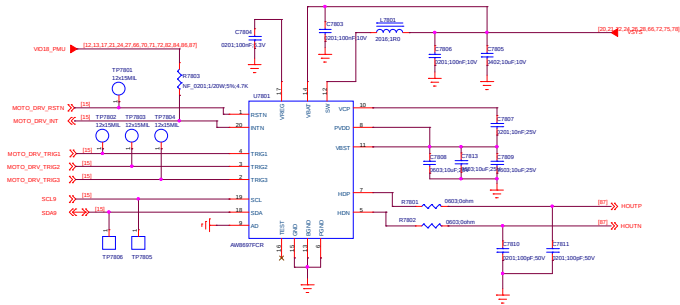
078_Flash/RGB/Vibrator

Flash LED



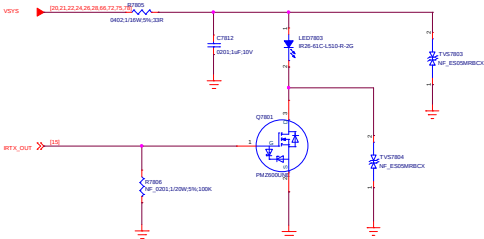
White LED

LRA Driver IC



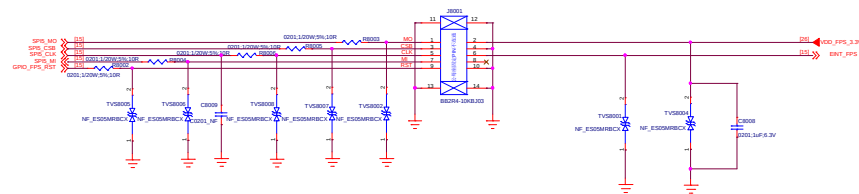
AW8695 I2C Address:
AD=0: 0xB4(W)/0xB5(R)
AD=1: 0xB6(W)/0xB7(R)

IR LED



080_FingerPrint

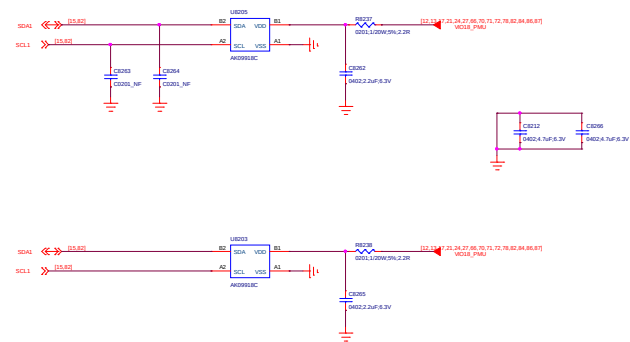
Fingerprint



Title 080_FingerPrint		REV: V1.0
DOCUMENT NO.: 00018_1_30_20210001_1440		Size D
DEPARTMENT: HR	DESIGNER: huangheng	
WINGTECH		
Date: Tuesday, August 10, 2021	Sheet 80 of 152	

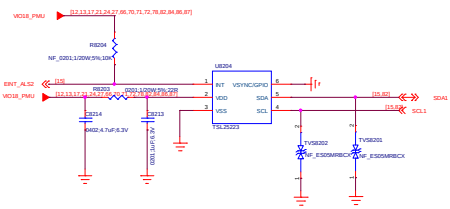
082_SENSORS

M-SENSOR

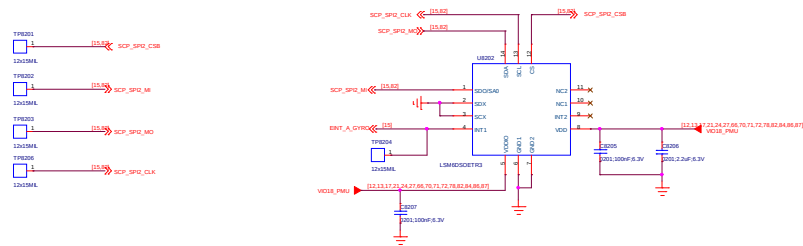


M-Sensor reserved

FRONT ALS SENSOR



Accerometer + GYRO SENSOR



REAR ALS SENSOR

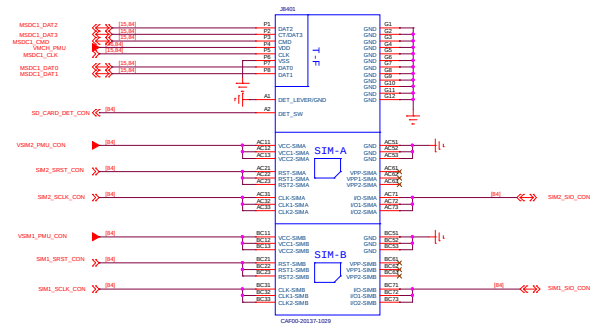
Schematic design notice of "82_PERI_SENSORS_MEMS_ALS/PS" page.

- Note 82-1: [M sensor] Keep a minimum distance of 15mm from power ICs / PCB traces of more than 100mA / magnet component. Check HW design notice for more detail
- Note 82-2: [A+G] For optimized GPS performance, please check HW design notice for Sensor selection guide
- Note 82-3: [A+G] MUST use SPI for optimized sensor hub performance
- Note 82-4: [A+G] Suggest choose sensor support FIFO watermark interrupt, otherwise we cannot support Hifi-sensor, daydream VR. And Sensor-location accuracy will become worse.
- Note 82-5: DO NOT share Sensor hub I2C to other non-SCP device

082_SENSORS		REV: V02
DOCUMENT NO.: 08024_1_16_20230802_1440	Size: D	
DEPARTMENT: 88	DESIGNER: huanghupeng	
Date: Tuesday, August 15, 2023		Sheet: 02 of 02

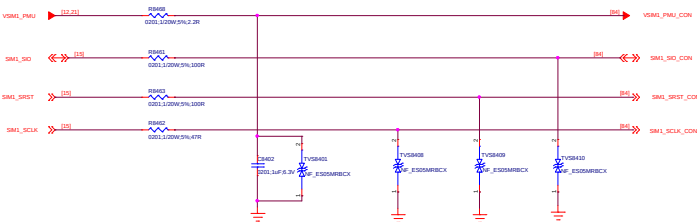
084_SIM/SD IF

CONNECTOR

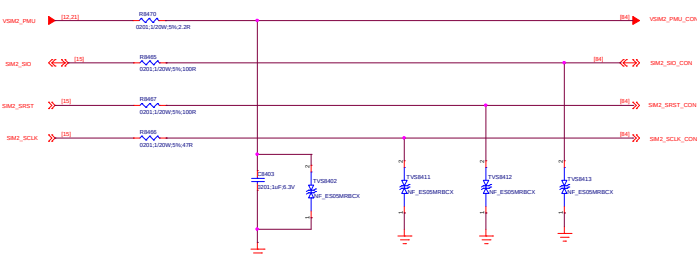


SIM CARD

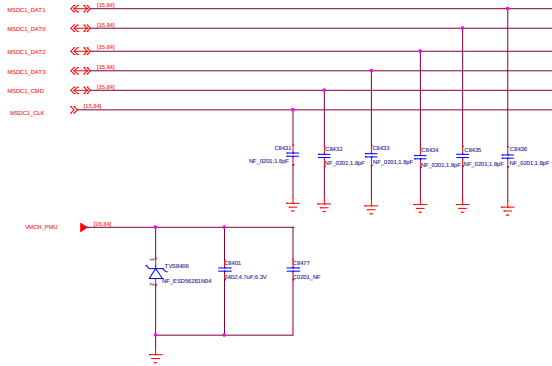
SIM1



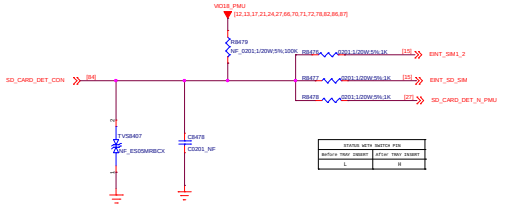
SIM2



T_Card



HOT PLUG DETECT



086_Testpoint/Shielding/GND

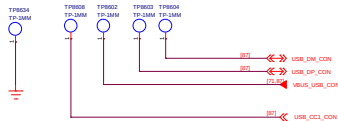
SIDE KEY



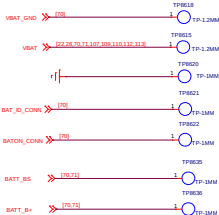
RAMDUMP



USB



Battery



UART

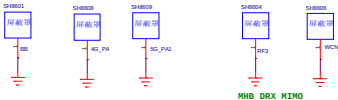


JTAG

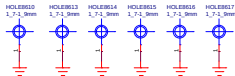


Shielding Cover

TOP



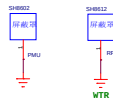
Hole



MIC LABLE & SN



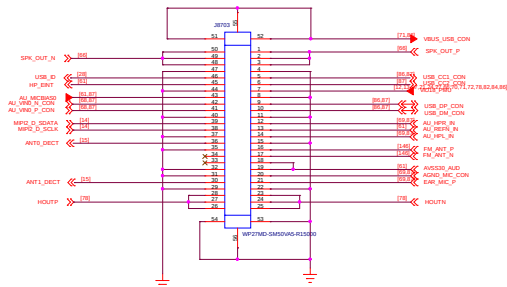
BOTTOM



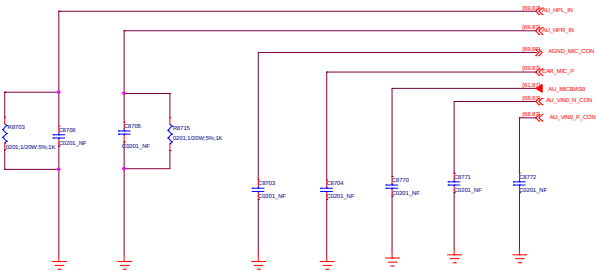
MARK



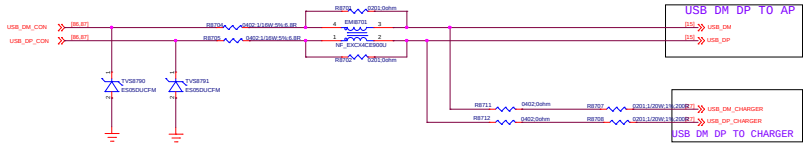
087_Sub PCB IF / USB IF



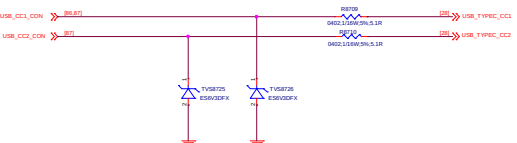
AUDIO CAP

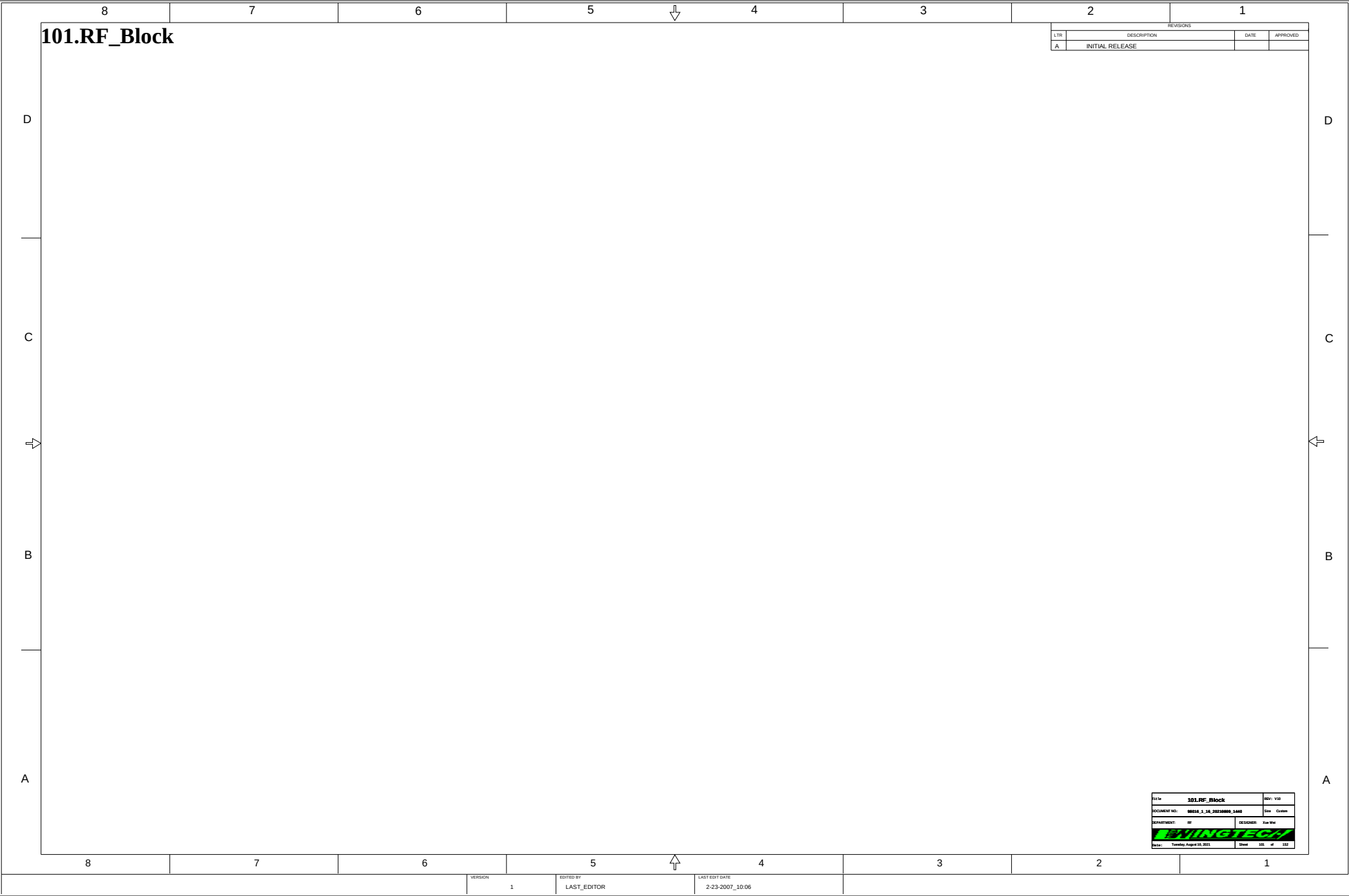


USB IF



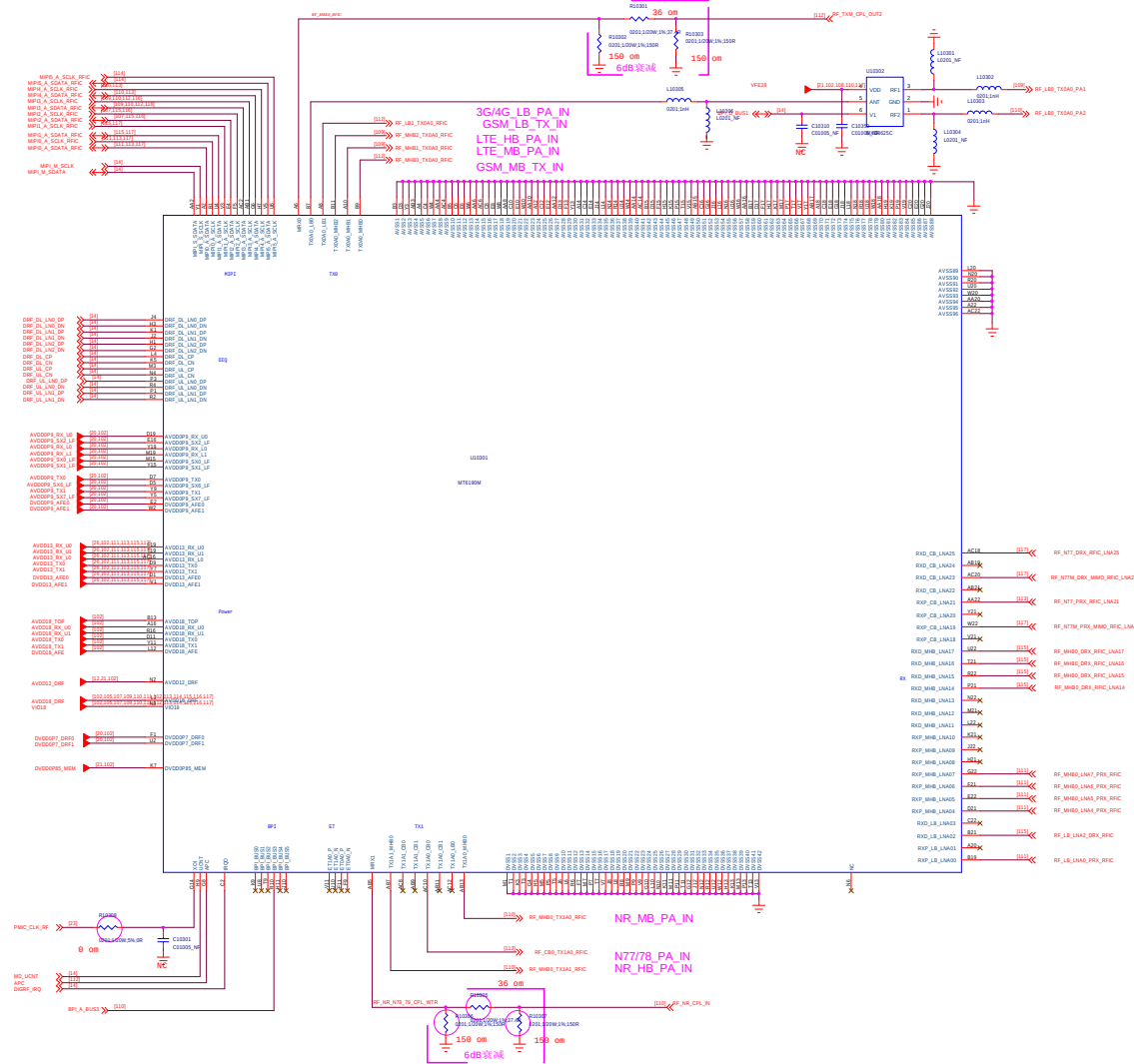
USB IF

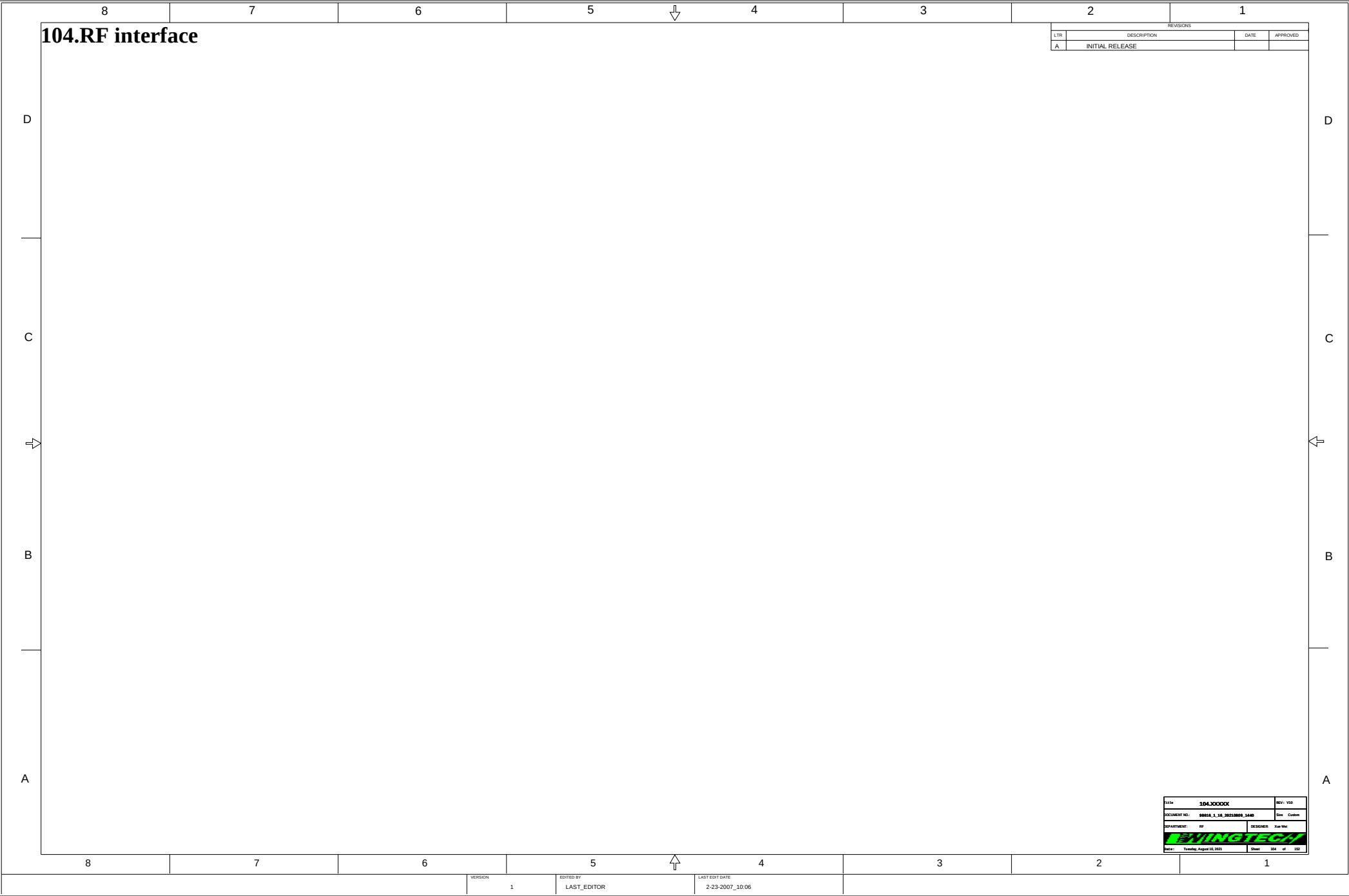




Transceiver MT6190M

Title		103.RF_Transceiver_MT619007		REV: V10	
DOCUMENT NO:				98016_1_16_20230809_1440	
DEPARTMENT:				RF	
DESIGNER:				Xue Wei	
					
Date:		Tuesday, August 10, 2021		Sheet 103 of 152	





104.RF interface

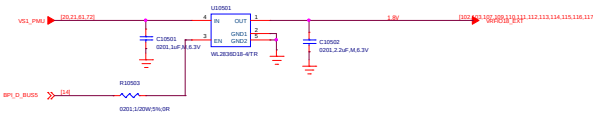
REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		

Title 104.XXXXX		REV: V01
DOCUMENT NO: 00016_1_16_20210000_1440		Size Custom
DEPARTMENT: RF		DESIGNER: Xue Wei
		
Date: Tuesday, August 10, 2021	Sheet 004 of 152	

VERSION	1	EDITED BY	LAST_EDITOR	LAST EDIT DATE	2-23-2007_10:06
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105_POWER_EXT

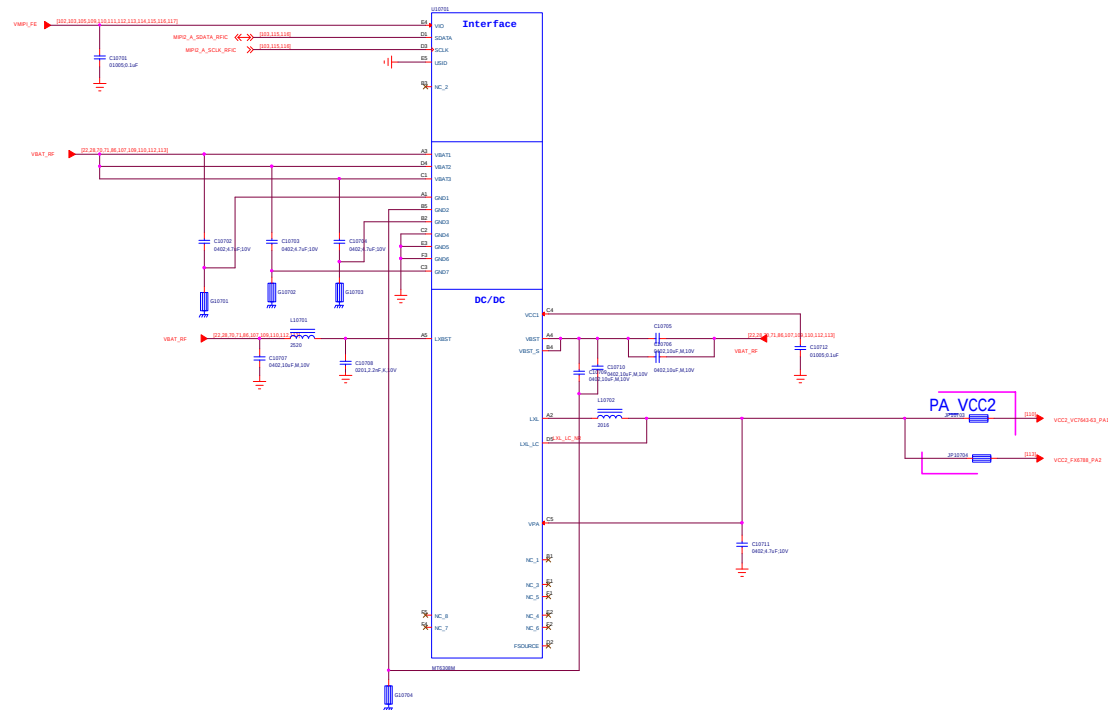
External VRFIO18_LDO



105_POWER_EXT		REV: V02
DOCUMENT NO: 00014_1_M_20210001_1408		SW: D
DEPARTMENT: EE	DESIGNER: Ning Xiong	
Date: Tuesday, August 10, 2021		Sheet: 005 of 100

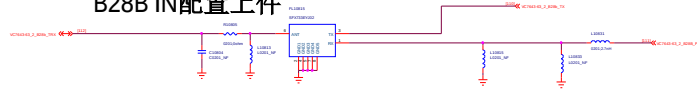


6308M_APT_2#

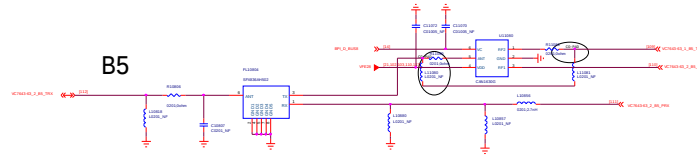


REV. 100		REV. 100	
107.APTIET_Module_2		107.APTIET_Module_2	
DOCUMENT NO. 00014_1_14_20230805_1400		Date: 08/05/2023	
DEPARTMENT: RF		DESIGNED: Xue Wen	
DATE: 2023-08-05		DRAWN: Xue Wen	
Checked: 2023-08-10		Sheet: 107 of 107	

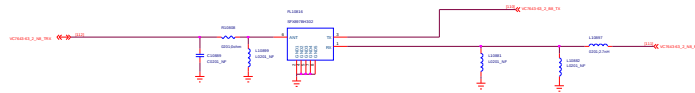
B28B IN配置上件



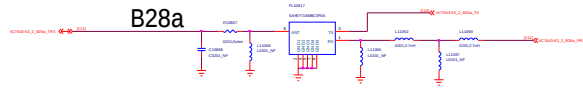
B5



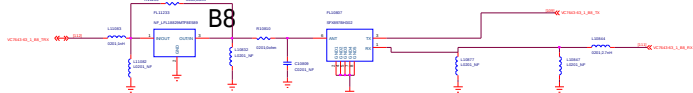
N8 CN/IN兼容设计, IN配置上件



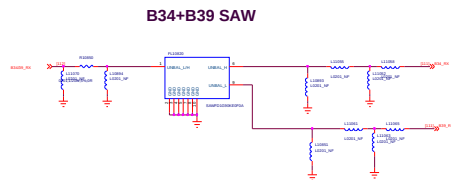
B28a



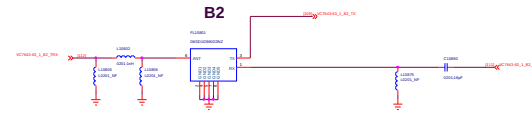
B8



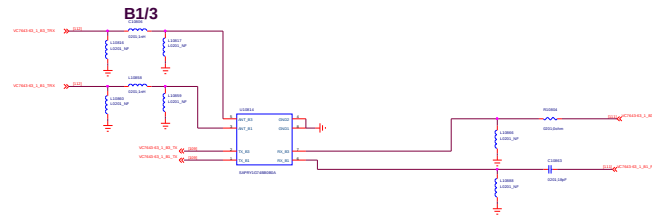
B34+B39 SAW



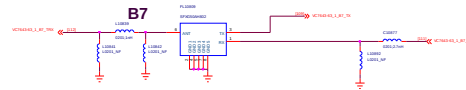
B2



B1/3

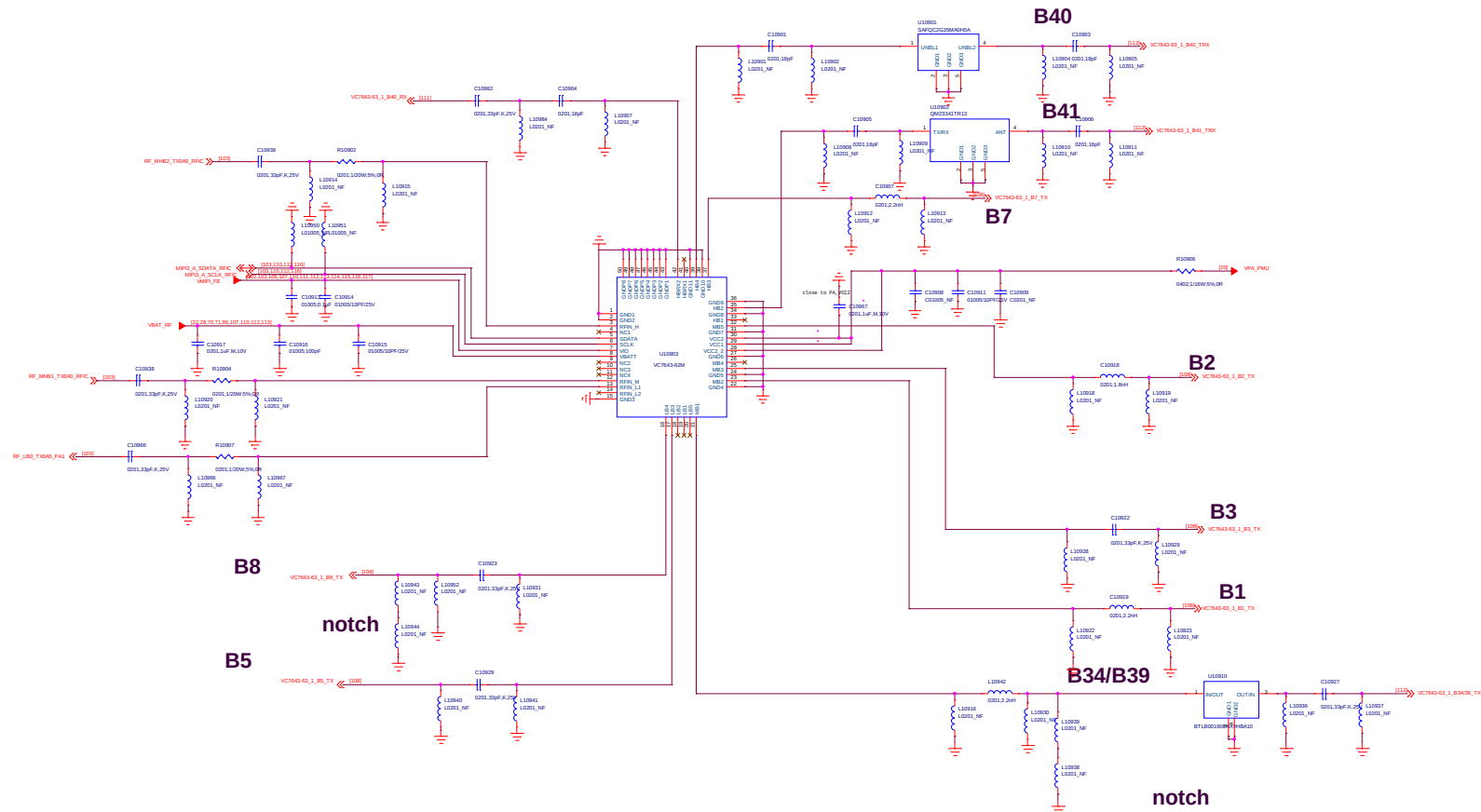


B7



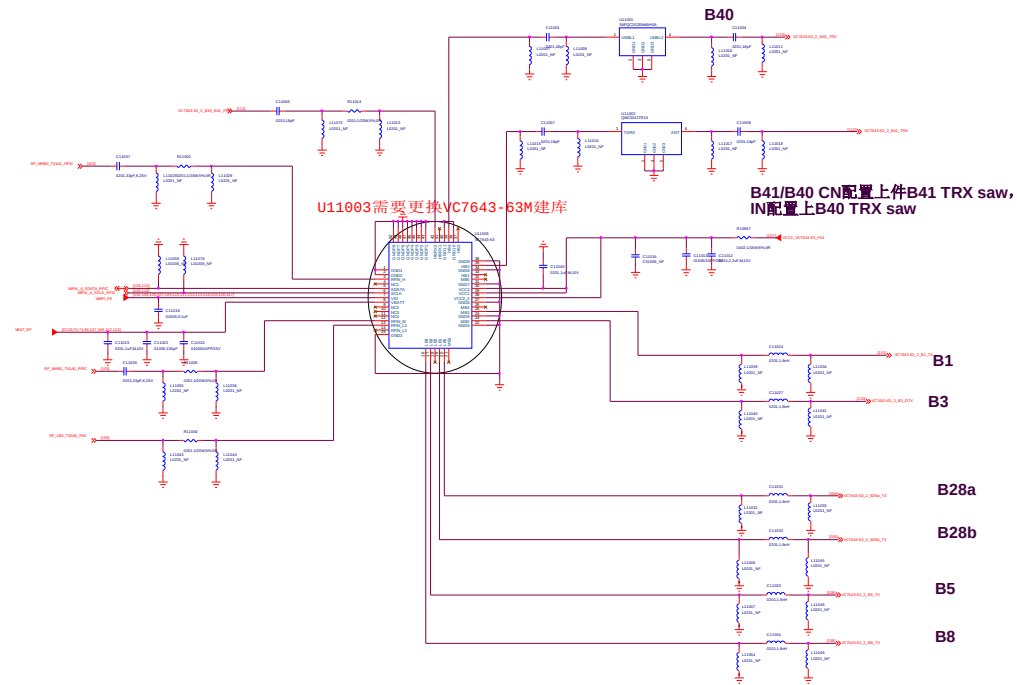
109.VC7643-62M #1

U10901需要更换SAFQC2G35MA0H0A建库

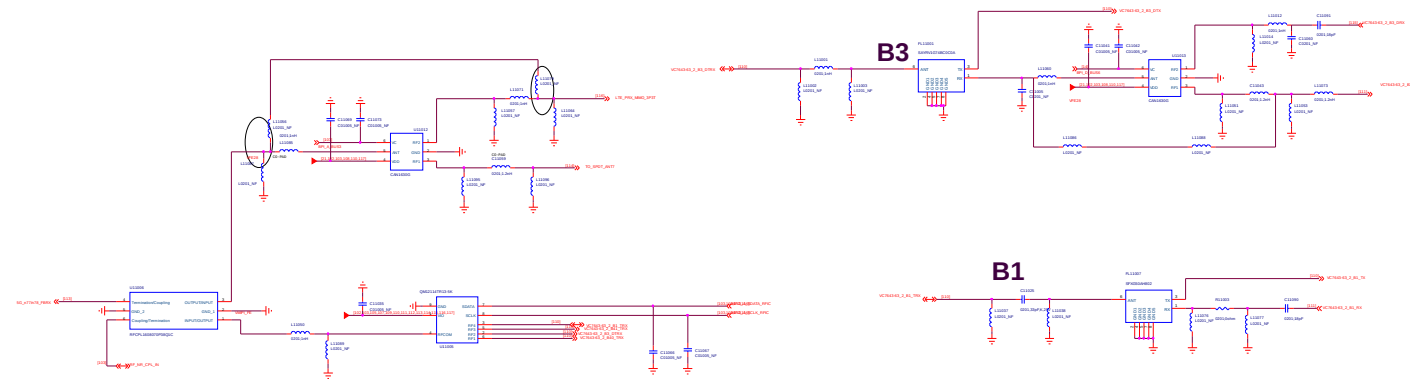


File	109.VC7643_62M#1	REV: V03
DOCUMENT NO:	88016_1_16_20210809_1440	Size: D
DEPARTMENT:	RF	DESIGNED: Xian-Yin
DATE:	Thursday, August 10, 2021	Sheet 109 of 120

110.VC7643-63#2

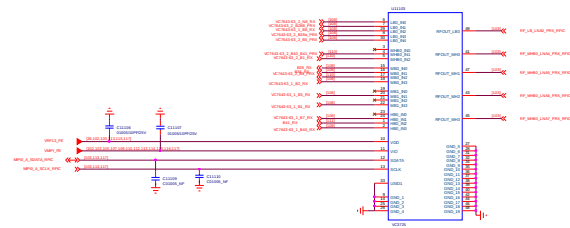
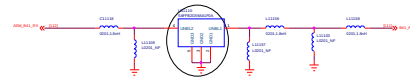


FL11001和FL11080器件需要更换一颗B1+B3 DPX:SAPRY1G74BB0B0A 库

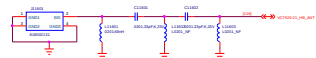


PRX_MXD83K5A

U11110需要更换SAFFB2G59AA1F0A库



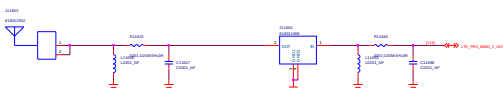
ANT0



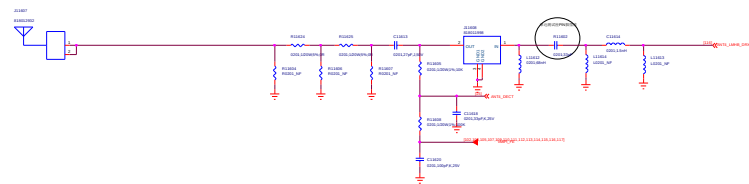
ANT1



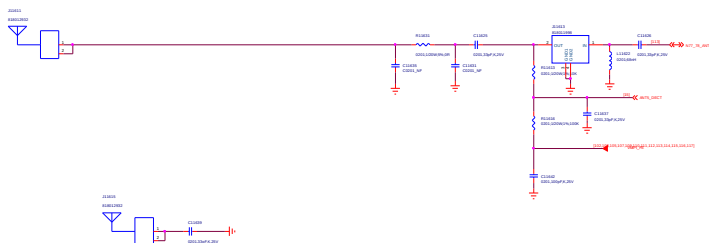
ANT2



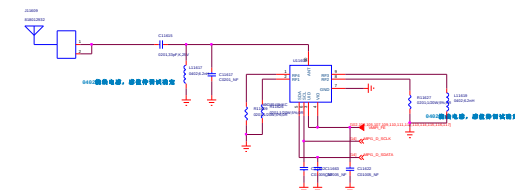
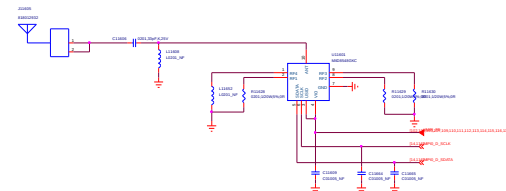
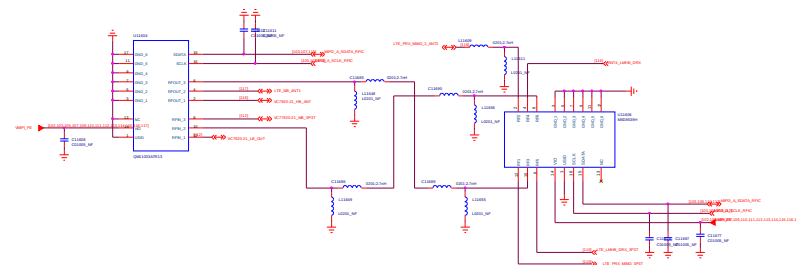
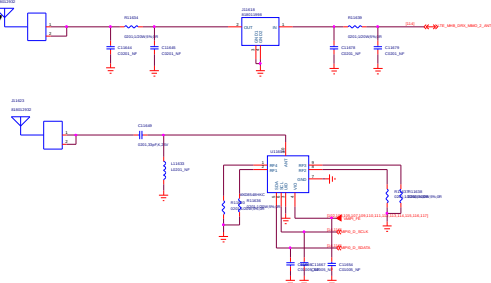
ANT4



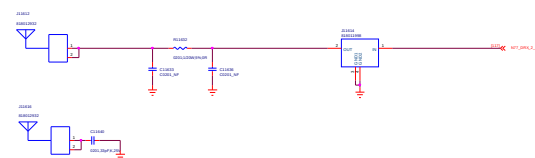
ANT5



ANT7



ANT3

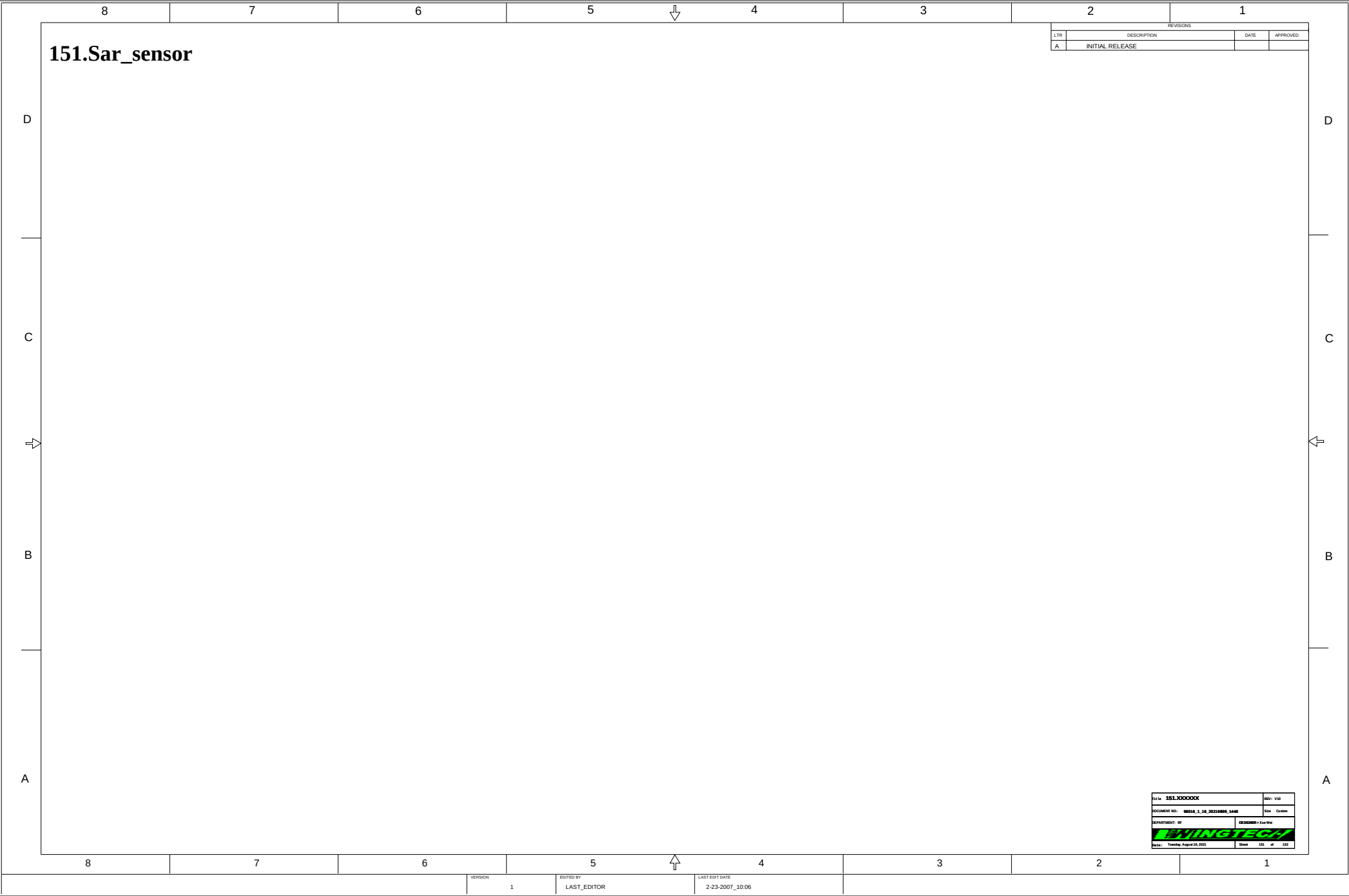


环型垫片



主板接地弹片





151.Sar_sensor

REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		

Part No.	151.000000	Rev.	100
DOCUMENT NO.	00010_1_16_20230000_1400	Date	Custom
DEPARTMENT: BY	DRAWING: BY		
WININGTECH		Sheet	101 of 102
Date: Tuesday, August 10, 2021			

